Maxima-Minima

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Maxima-Minima

(A) GENERAL INTRODUCTION :

The notion of optimising functions is one of the most useful application of calculus used in almost every sphere of life including geometry, business, trade, industries, economics, medicines and even at home. In this chapter we shall see that calculus defines the notion of maxima and minima and distinguishes it form the greatest and least value of global maxima and global minima of a function.

(B) DEFINITION OF MAXIMA & MINIMA :

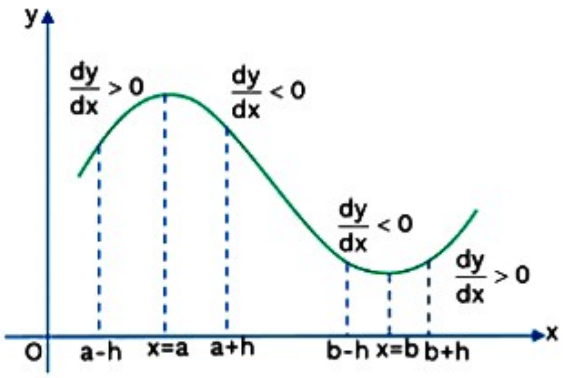
Symbolically, $\left.\begin{matrix} f(a) > f(a+h) \\ f(a) > f(a-h) \end{matrix}\right] \Rightarrow x = a$ gives maxima for a sufficiently small positive h.

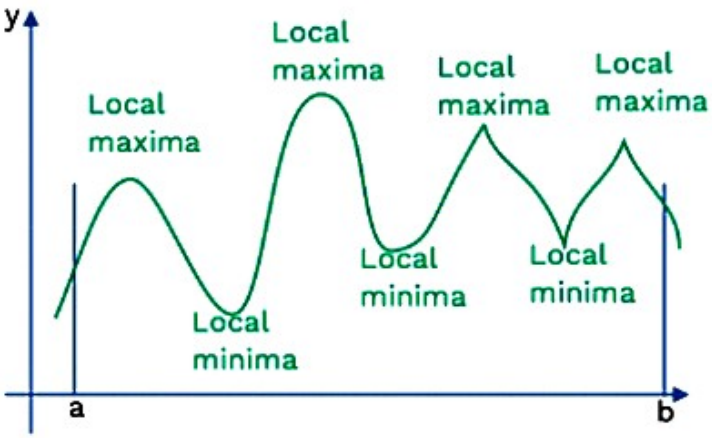
Similarly, a function f(x) is said to have a minimum value at x = b if f(b) is lesser than other value assumed by f(x) in the immediate neighbourhood at x = b. Symbolically if

$\left.\begin{matrix} f(b) < f(b+h) \\ f(b) < f(b-h) \end{matrix}\right] \Rightarrow x = b$ gives minima for a sufficiently small positive h.

Definition

A function f(x) is said to have maximum at x = a if f(a) is greater than every other value assumed by f(x) in the immediate neighbourhood of x = a.





Points to Remember!!!

(i) The maximum and minimum values of a function are also known as local/ relative maxima or local / relative minima as these are the greatest and least values of the function relative to some neighbourhood of the point in question.

Test for local maximum / minimum, when f(x) is differentiable :

(1) First-order derivative test in ascertaining the maxima or minima :

Consider the interval (a–h, a), we find f(x) is increasing $\Rightarrow \frac{dy}{dx} > 0$. Similarly, for the interval (a, a+h), we find f(x) is decreasing $\Rightarrow \frac{dy}{dx} < 0$. Hence, at the point x = a (maxima); $\frac{dy}{dx} = 0$.

Similarly, $\frac{dy}{dx} = 0$ at x = b which is the point of minima.

Hence $\frac{dy}{dx} = 0$ is the necessary condition for maxima or minima.

These points, where $\frac{dy}{dx}$ vanishes, are known as **Stationary Points** as instantaneous rate of function momentarily ceases at this point.

Similarly, $\left.\begin{matrix} f'(a-h) > 0 \\ f'(a+h) < 0 \end{matrix}\right] \Rightarrow x = a$ is a point of local maxima, where f'(a) = 0. It means that f'(x) should change its sign from positive to negative.

However, if f'(x) does not change sign, i.e., has the same sign in a certain complete neighbourhood of c, then f(x) is either increasing or decreasing throughout implying that f(c) is not an extreme value of f, e.g., $f(x) = x^3$ at x = 0.

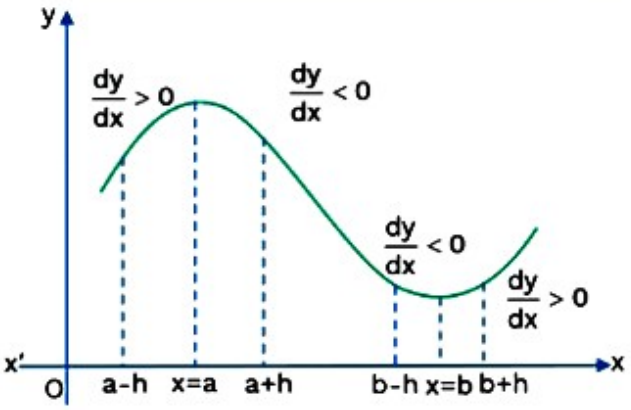
Points to Remember!!!

(ii) The term 'extremum' or (extrema) or 'turning value' is used both for maximum or a minimum value.

(iii) The maximum (minimum) value of a function may not be the greatest (least) value in a finite interval.

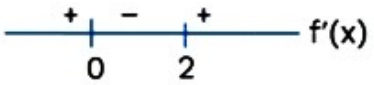
(iv) A function can have several maximum & minimum values and a minimum value may even be greater than a maximum value.

(v) Maximum and minimum values of a continuous function occur alternately & between two consecutive minimum values there is a maximum value & vice versa.



Q.1 **Let $f(x) = x^3 - 3x^2 + 6$, find the point at which f(x) assumes local maximum and local minimum.**

Sol. $f'(x) = 3x^2 - 6x$
$= 3x(x - 2)$



$\therefore$ f(x) has local maxima at x = 0 and local minima at x = 2
Local maxima = f(0) = 6
Local minima = f(2) = 2

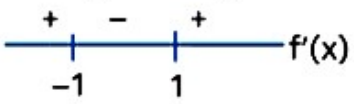
Q.2 **Let $f(x) = x^3$. Find the point at which f(x) assumes local maxima and local minima.**

Sol. $f'(x) = 3x^2 \geq 0$

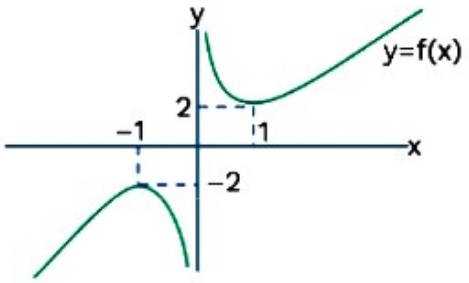
So, f(x) is always monotonic increasing.
$\therefore$ there is no local maxima or minima.

Q.3 **Let $f(x) = x + \frac{1}{x}, x \neq 0$ .Discuss the maximum and minimum values of f(x).**

Sol. $f'(x) = 1 - \frac{1}{x^2} = \frac{(x-1)(x+1)}{x^2}$



So, f(x) will have local maxima at
x = −1 and local minima at x = 1
Local maxima = f(−1) = −2 and
Local minima = f(1) = 2.



Q.4 **Find the local maximum and local minima of $f(x) = x^3 + 3x$ in [−2, 4].**

Sol. $f'(x) = 3x^2 + 3 > 0 \forall x \in R$

So, f(x) is always increasing.
So, $f_{min} = f(-2) = -14$

$f_{max} = f(4) = 76$.

Q.5 If $f(x)=\begin{cases}3x^2+12x-1, & -1\le x\le 2\\ 37-x, & 2<x\le 3\end{cases}$, **then**

(A) f(x) is increasing on [–1, 2] **(B) f(x) is continuous on [–1, 3]**
(C) f'(x) does not exist at x = 2. **(D) f(x) has the maximum value at x = 2**

Sol. **(ABCD)**

$$f'(x)=\begin{cases}6x+12, & -1\le x\le 2\\ -1, & 2<x\le 3\end{cases}$$

So $f'(x)>0\forall x\in[-1,2]$ and $f'(x)<0\forall x\in(2,3]$

Also, f(x) attains maximum value at x = 2.

$f'(2^-)=24;\quad f'(2^+)=-1$

$\therefore$ f'(x) does not exist at x = 2.

$f(2)=35=f(2^-)=f(2^+)$

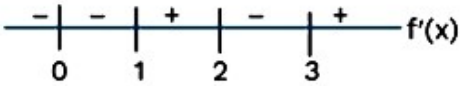
So, all the options are correct.

Q.6 **The function** $f(x)=\int_{-1}^{x} t(e^t-1)(t-1)(t-2)^3(t-3)^5\,dt$ **has a local maximum at x equal to**

(A) 0 **(B) 1** **(C) 2** **(D) 3**

Sol. **(C)**

$f'(x)=x(e^x-1)(x-1)(x-2)^3(x-3)^5$



Since at maxima, f'(x) changes from positive to negative, f(x) has a local maxima at x = 2.

Q.7 **Find the set of critical points of the function**

$$f(x)=x-\log x+\int_2^x\left(\frac{1}{z}-2-2\cos 4z\right)dz.$$

Sol. $f'(x)=1-\frac{1}{x}+\frac{1}{x}-2-2\cos 4x$

$=-1-2\cos 4x$

$\Rightarrow f'(x)=0\Rightarrow\cos 4x=\frac{-1}{2}$

$$\Rightarrow 4x = 2n\pi \pm \frac{2\pi}{3}, n \in I$$

$$\Rightarrow x = \frac{n\pi}{2} \pm \frac{\pi}{6}, n \in I$$

$\Rightarrow$ Set of critical points is $x \in \left\{\frac{\pi}{6}\right\} \cup \left(\frac{n\pi}{2} \pm \frac{\pi}{6}\right), n \in N \quad (x > 0)$

Q.8 **Let f(x) = sin x – x on $\left[0, \frac{\pi}{2}\right]$, find local maximum and local minimum.**

Sol. $f'(x) = \cos x - 1 \qquad \leq 0 \ \forall \ x \in \left[0, \frac{\pi}{2}\right]$

$f'(0) = 0.$

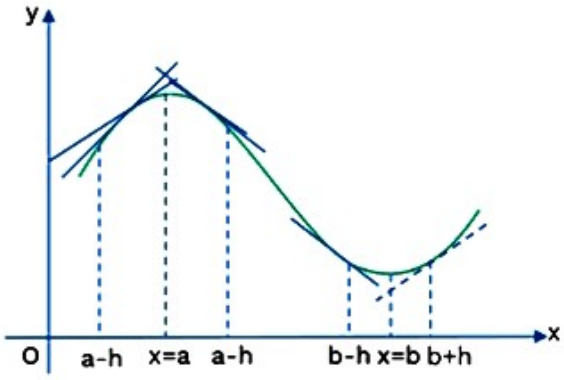
$\therefore$ f(x) is decreasing on $\left[0, \frac{\pi}{2}\right]$

$$\Rightarrow f_{min} = f\left(\frac{\pi}{2}\right) = 1 - \frac{\pi}{2}$$

$$\Rightarrow f_{max} = f(0) = 0$$

(2) Use of second order derivative in ascertaining the Maxima or Minima for a differentiable function :

As shown in the figure it is clear that as x increases from a-h to a + h, the function $\frac{dy}{dx}$ continuously decreases, i.e. (+)ve for x < a, zero at x = a and (–)ve for x > a. Hence $\frac{dy}{dx}$ itself is a decreasing function. Therefore $\frac{d^2y}{dx^2} < 0$ in (a – h, a + h).



Hence at local maxima, $\frac{dy}{dx} = 0$ and $\frac{d^2y}{dx^2} < 0$.

f'(a) = 0 and f''(a) < 0

Similarly at local minima, $\frac{dy}{dx} = 0$ and $\frac{d^2y}{dx^2} > 0$

i.e. f'(b) = 0 and f''(b) > 0

Hence if

(a) f(a) is a maximum value of the function f, then $f'(a) = 0$ and $f''(a) < 0$

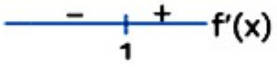
(b) f(b) is minimum value of function f, then $f'(b) = 0$ and $f''(b) > 0$.

However, if $f''(c) = 0$ then the test fails. In this case f can still have a maxima or minima or point of inflection (neither maxima nor minima). In this case revert back to the first order derivative check for ascertaining the maxima or minima.

Q. Let $f(x) = (x - 1)^4$. Discuss the point at which f(x) assumes the minimum value.

Sol. $f'(x) = 4(x - 1)^3$

$f''(x) = 12(x - 1)^2$



Since f'(x) changes from negative to positive at x = 1, f(x) attains minima at x = 1.

$\Rightarrow f_{min} = f(1) = 0$

Also, $f''(x) \geq 0$

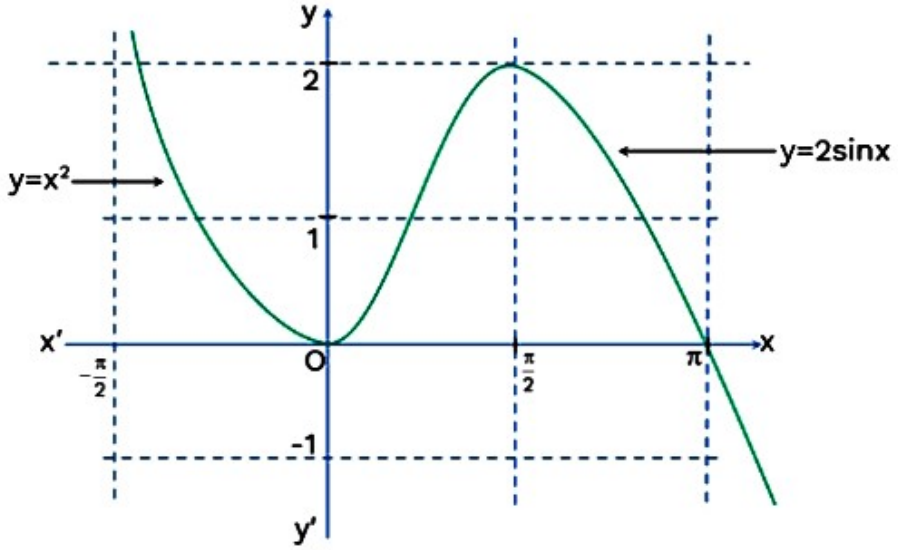
$\therefore$ Graph will be concave up

(3) n^{th} Derivative Test :

It is nothing but the general version of the second derivative test. It says that if $f'(a) = f''(a) = f'''(a) \ldots f^n(a) = 0$ and $f^{n+1}(a) \neq 0$ (all derivatives does not vanish and (n + 1)th order derivative does not vanish at x = a), then f(x) would have a local maximum or minimum at x = a if n is odd natural number and that x = a would be a point of local maxima if $f^{n+1}(a) < 0$and would be a point of local minima if $f^{n+1}(a) > 0$. However, if n is even, then f has neither a maxima nor a minima at x = a.

Q. **If $f(x)=\begin{cases}x^2 & ,x\le 0\\ 2\sin x, & x>0\end{cases}$, investigate the function at x = 0 for maxima/minima.**

Sol. Analyzing the graph of f(x), we get x = 0 is a point of minima.



Q. **The function $y=\dfrac{ax+b}{(x-1)(x-4)}$ has turning point at P(2, –1). Then find the value of a and b.**

Sol. $y=\dfrac{ax+b}{(x-1)(x-4)}=\dfrac{ax+b}{x^2-5x+4}$ has turning point at P(2, –1)

$\Rightarrow$ P(2, – 1) lies on the curve $\Rightarrow$ 2a + b = 2 ...(i)

Also $\dfrac{dy}{dx}=0$ at $P(2,-1)$

Now $\dfrac{dy}{dx}=\dfrac{a(x^2-5x+4)-(2x-5)(ax+b)}{(x^2-5x+4)^2}$

At $P(2,-1), \dfrac{dy}{dx}=\dfrac{-2a+2a+b}{4}=0$

$\Rightarrow$ b = 0 $\Rightarrow$ a = 1 [from equation (i)]

Q. **Find the points of maxima and minima of the function $f(x)=12x^5-45x^4+40x^3+40$. Check whether second derivative can be used to find the point of extrema.**

Sol. We have $f(x)=12x^5-45x^4+40x^3+40$

$f'(x) = 60x^4 - 180x^3 + 120x^2 = 60x^2(x-1)(x-2)$ and $f''(x) = 60(4x^3 - 9x^2 + 4x)$

The critical points of $f(x)$ are $x = 0, 1, 2$

At the critical points, we have

$f''(0) = 0 \Rightarrow$ more investigation required

$f''(1) = 60(4 - 9 + 4) < 0 \Rightarrow$ maxima at $x = 1$.

$f''(2) = 60(32 - 36 + 8) > 0 \Rightarrow$ minima at $x = 2$.

Hence, we can see that the nature of the critical point at $x = 0$ cannot be predicted by the use of second derivative.

Since $f'(x)$ does not change sign as x passes through 0, hence $x = 0$ is not an extrema.

Q. **Discus the extremum of** $f(x) = x^2 + \frac{1}{x^2}$.

Sol. $f(x) = x^2 + \frac{1}{x^2}$

$f'(x) = 2x - \frac{2}{x^3}$

Let $f'(x) = 0 \Rightarrow x^4 = 1 \Rightarrow x = \pm 1$

Also, $f''(x) = 2 + \frac{6}{x^4} > 0$ for all $x \neq 0$

$\Rightarrow$ Both the point $x = 1$ and $x = -1$ are the points of minima.

Q. **The function** $f(x) = (4\sin^2 x - 1)^n (x^2 - x + 1), n \in N$, **has a local minimum at** $x = \frac{\pi}{6}$, **then**

(A) n is any even number **(B) n is an odd number**

(C) n is odd prime number **(D) n is natural number**

Sol. **(A)**

$$f(x) = \left(4\sin^2 x - 1\right)^n \left(x^2 - x + 1\right)$$

$x^2 - x + 1 > 0 \; \forall \; x \in R$

$$f\left(\frac{\pi}{6}\right) = 0 \Rightarrow f\left(\frac{\pi^+}{6}\right) = \lim_{x \to \frac{\pi^+}{6}} \left(4\sin^2 x - 1\right)^n \left(x^2 - x + 1\right) \to 0^+$$

$$f\left(\frac{\pi}{6}\right) = 0 \Rightarrow f\left(\frac{\pi^-}{6}\right) = \lim_{x \to \frac{\pi^-}{6}} \left(4\sin^2 x - 1\right)^n \left(x^2 - x + 1\right) \to \left(0^-\right)^n$$

(a positive value)

$f\left(\frac{\pi^-}{6}\right) > 0$ if n is an even number.

When F(x) is not differentiable at x = a :

Case-I :

When f(x) is continuous at x = a and f'(a – h) and f'(a + h) exist and are non-zero, then f(x) has a local maximum or minimum at x = a if f'(a – h) and f'(a + h) are of opposite signs.

If f'(a – h) > 0 and f'(a + h) < 0, then x = a will be a point of local maximum.

If f'(a – h) < 0 and f'(a + h) > 0, then x = a will be point of local minimum.

Case-II :

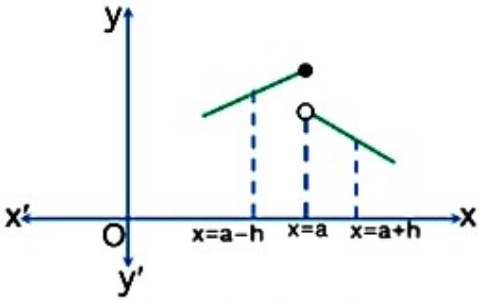
When f(x) is continuous and f'(a – h) and f'(a + h) exist but one of them is zero, we should infer the information about the existence of local maxima/minima from the basic definition of local maxima/minima.

Case-III :

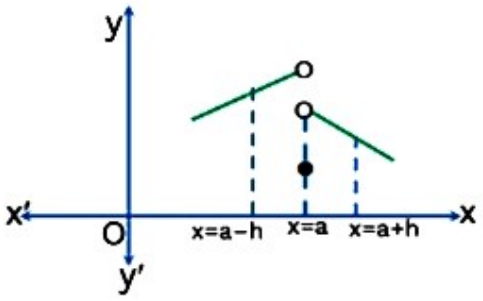
If f(x) is not continuous at x = a and f'(a – h) and / or f'(a + h) are not finite, then compare the values of f(x) at the neighbouring points of x = a.

It is advisable to draw the graph of the function in the vicinity of the point x = a, because the graph would give us the clear picture about the existence of local maxima/ minima at x = a.

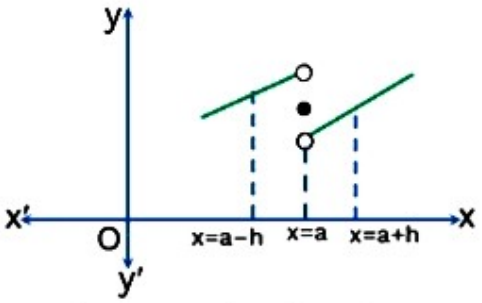
Consider the following cases :



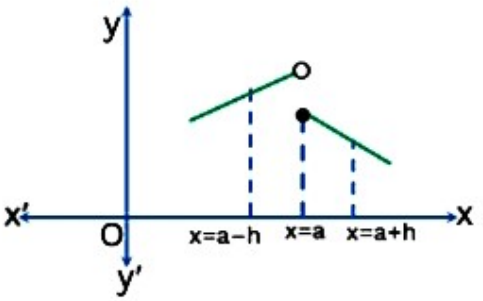
x = a is point of maxima as
f(a) > f(a−h) and f(a) > f(a + h)



x = a is point of minima as
f(a) < f(a−h) and f(a) < f(a + h)



x = a is not a point of maxima
or minima as f(a) < f(a−h) and f(a) > f(a + h)



x = a is not a point of maxima or minima
as f(a) < f(a−h) and f(a) > f(a + h)

Concept of global maximum/minimum :

Let y = f(x) be a given function with domain D.
Let $[a,b] \subseteq D$.

Global maximum and minimum in [a, b] would occur at critical points of f(x) within [a, b] or at the endpoints of the interval.

Global maximum/minimum in [a, b]

In order to find the global maximum and minimum of f(x) in [a, b], find the critical points of f(x) in (a, b). Let $c_1, c_2, \ldots c_n$ be the different critical points. Find the value of the function at these critical points. Let $f(c_1), f(c_2), \ldots f(c_n)$ be the values of the function at critical points.

Say, $M_1 = \max\{f(a), f(c_1), f(c_2), \ldots f(c_n), f(b)\}$

and $M_2 = \min\{f(a), f(c_1), f(c_2), \ldots f(c_n), f(b)\}$

Then M_1 is the greatest value of f(x) in [a, b]
and M_2 is the least value of f(x) in [a, b].

Q. **Find the maximum value of** $f(x) = \left(\frac{1}{x}\right)^x$

Sol. $f(x) = \left(\frac{1}{x}\right)^x \Rightarrow f'(x) = \left(\frac{1}{x}\right)^x \left(\ln\frac{1}{x} - 1\right)$

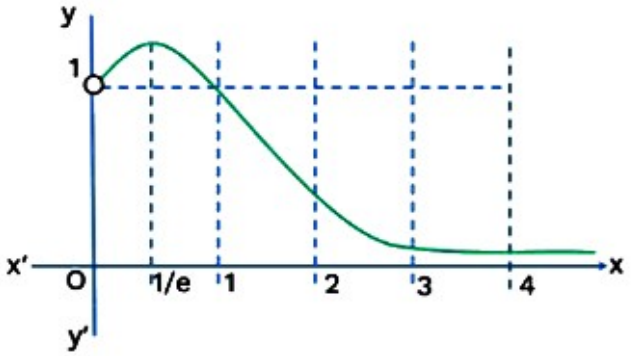
$f'(x) = 0 \Rightarrow \ln\frac{1}{x} = 1 \Rightarrow \frac{1}{x} = e \Rightarrow x = \frac{1}{e}$

Also for $x < \frac{1}{e}, f'(x)$ is positive and for

$x > \frac{1}{e}$, $f'(x)$ is negative.

Hence, $x = \frac{1}{e}$ is point of maxima.

Therefore, the maximum value of function is $e^{\frac{1}{e}}$.



Q. **Let $f(x) = 2x^3 - 9x^2 + 12x + 6$. Discuss the global maxima and minima of f(x) in [0, 2] and (1, 3).**

Sol. $f(x) = 2x^3 - 9x^2 + 12x + 6$

$f'(x) = 6x^2 - 18x + 12 = 6(x^2 - 3x + 2) = 6(x - 1)(x - 2)$

Clearly the critical points of f(x) in [0, 2] is x = 1 and x = 2.

$f(1) > f(2) > f(0)$

Thus x = 0 is the point of global minima of f(x) in [0, 2] and x = 1 is the point of global maximum.

For $x \in (1, 3)$

Clearly c = 2 Is the critical point in (1, 3)

$f(2) = 10$. $\lim_{x \to 1^+} f(x) = 11$ and $\lim_{x \to 3^-} f(x) = 15$

Thus x = 2 is the point of global minimum and global maximum in (1, 3) does not exist.

Q. **Find the greatest and least value of function $f(x) = 3x^4 - 8x^3 - 18x^2 + 1$.**

Sol. We have

$$f(x) = 3x^4 - 8x^3 - 18x^2 + 1$$

and $$f'(x) = 12x^3 - 24x^2 - 36x = 12x(x+1)(x-3).$$

The points at which f(x) may have extreme value, are the critical points x = –1, 0, 3

Now; $f(-1) = -6, f(0) = 1, f(3) = -134$, and $f(\pm\infty) \to +\infty$.

Hence, the least value of the function is –134 whereas the greatest value does not exist.

Q. **Find the greatest and least value of function** $f(x) = \begin{cases} -x, & -1 \le x < 0 \\ 2-(x-1)^2, & 0 \le x \le 2 \end{cases}$.

Sol. We have $f(x) = \begin{cases} -x, & -1 \le x < 0 \\ 2-(x-1)^2, & 0 \le x \le 2 \end{cases}$ and $f'(x) = \begin{cases} -1, & -1 < x < 0 \\ -2(x-1), & 0 \le x < 2 \end{cases}$

Thus, the points at which f(x) may have extreme values, are the critical points x = 0, 1 $\left[f'(x) = 0,\ f'(x) = \text{DNE}\right]$

and the endpoints x = –1, 2

Now; $f(-1) = 1, f(1) = 2$ and $f(2) = 1$.

Since f is discontinuous at x = 0,

as $f(0^-) \to 0, f(0^+) \to 1$ and $f(0) = 1$

The largest and the smallest among the above six values are 2 and 0 respectively. Hence, the greatest value is 2 but the least value does not exist since the function approaches but is never equal to 0.

Q. **Find greatest and least values of** $f(x) = \dfrac{a^2}{x} + \dfrac{b^2}{1-x}, x \in (0,1)(a, b > 0)$.

Sol. We have

$$f(x) = \frac{a^2}{x} + \frac{b^2}{1-x}, x \in (0,1)$$

and $$f'(x) = \frac{-a^2}{x^2} + \frac{b^2}{(1-x)^2}$$

which exist everywhere in (0, 1) and vanishes at points, given by

$$\frac{b^2}{(1-x)^2} = \frac{a^2}{x^2}$$

$$a^2(1-x)^2 = b^2x^2$$

i.e. $a(1-x) = bx$ i.e. $x = \frac{a}{a+b}$

To find the greatest and least value, we need to check the values of f(x) at $x = \frac{a}{a+b}$.

$$f\left(\frac{a}{a+b}\right) = (a+b)^2$$

Hence, we have
Least value = $(a + b)^2$
And greatest value does not exist.

Q. **Find greatest and least values of** $f(x) = \frac{(a+x)(b+x)}{(c+x)}, x > -c$

Sol. We have $f(x) = \frac{(a+x)(b+x)}{(c+x)}, x > -c$

and $f'(x) = \frac{(c+x)(2x+a+b) - \left[x^2 + (a+b)x + ab\right]}{(c+x)^2}$

$$= \frac{x^2 + 2cx + ac + bc - ab}{(c+x)^2}, x \in (-c, \infty)$$

which vanishes at points given by
$x^2 + 2cx + ac + bc - ab = 0$

i.e. $x = -c \pm \sqrt{c^2 - (ac + bc - ab)} = -c \pm \sqrt{(a-c)(b-c)}$

Thus, the expression for f'(x) can be written as

$$f'(x) = \frac{(x-\alpha)(x-\beta)}{(c+x)^2}$$

Choosing $\alpha = -c - \sqrt{(a-c)(b-c)}$ and $\beta = -c + \sqrt{(a-c)(b-c)}$

The critical point $x = \alpha$ is of no interest since it does lie in the interval $(-c, \infty)$.

Now; we have

$f(-c^+) \to \infty, f(\infty) \to \infty$

and $f(\beta) = \dfrac{\left(a-c+\sqrt{(a-c)(b-c)}\right)\left(b-c+\sqrt{(a-c)(b-c)}\right)}{c-c+\sqrt{(a-c)(b-c)}}$

$$= \frac{(a-c)(b-c)+(a+b-2c)\sqrt{(a-c)(b-c)}+(a-c)(b-c)}{\sqrt{(a-c)(b-c)}}$$

$$= 2\sqrt{(a-c)(b-c)}+a+b-2c$$

$$= a-c+b-c+2\sqrt{(a-c)(b-c)}$$

$$= \left(\sqrt{(a-c)}+\sqrt{(b-c)}\right)^2$$

Hence, we have

Least value $= \left(\sqrt{(a-c)}+\sqrt{(b-c)}\right)^2$ and greatest value does not exist.

Q.1 **Discuss the global maxima and global minima of $f(x) = \tan^{-1}x - \log_e x$ in $\left[\frac{1}{\sqrt{3}}, \sqrt{3}\right]$.**

Sol. $f'(x) = \dfrac{1}{1+x^2} - \dfrac{1}{x}$

$$= \frac{-\left(x^2-x+1\right)}{x\left(1+x^2\right)}$$

$\Rightarrow f'(x) < 0 \ \forall x \in \left[\frac{1}{\sqrt{3}}, \sqrt{3}\right]$

$\Rightarrow f(x)$ is decreasing monotonically.

$\Rightarrow$ Maximum and Minimum values will be at end points.

$\Rightarrow f_{max} = f\left(\frac{1}{\sqrt{3}}\right) = \frac{\pi}{6} + \frac{\ell n 3}{2}$

$\Rightarrow f_{min} = f\left(\sqrt{3}\right) = \frac{\pi}{3} - \frac{\ell n 3}{2}$

Q.2 **Find the range of function $f(x) = 2\sqrt{x-2} + \sqrt{4-x}$.**

Sol. $f'(x) = \frac{1}{\sqrt{x-2}} - \frac{1}{2\sqrt{4-x}}$

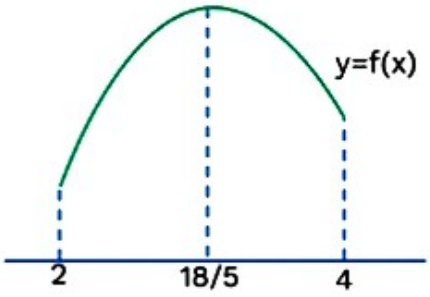
$f'(x) > 0 \Rightarrow \frac{1}{\sqrt{x-2}} > \frac{1}{2\sqrt{4-x}}$

$\Rightarrow 2\sqrt{4-x} > \sqrt{x-2}$

$\Rightarrow 16 - 4x > x - 2$

$\Rightarrow 18 > 5x$

$\Rightarrow x < \frac{18}{5}$



Also, domain of f(x) is [2, 4]

So, f(x) increases in $\left[2, \frac{18}{5}\right]$ and then

decreases on $\left(\frac{18}{5}, 4\right]$

$f_{max} = f\left(\frac{18}{5}\right) = 4\sqrt{\frac{2}{5}} + \sqrt{\frac{2}{5}} = \sqrt{10}$

$f_{min} = f(2) = \sqrt{2}$

$\Rightarrow$ Range is $\left[\sqrt{2}, \sqrt{10}\right]$.

Q.3 **Find the difference between the greatest and least value of the function f(x) = sin2x – x on $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$.**

Sol. $f'(x) = 2\cos 2x - 1$

$f'(x) = 0 \Rightarrow \cos 2x = \frac{1}{2}$

$\Rightarrow 2x = 2n\pi \pm \frac{\pi}{3}$

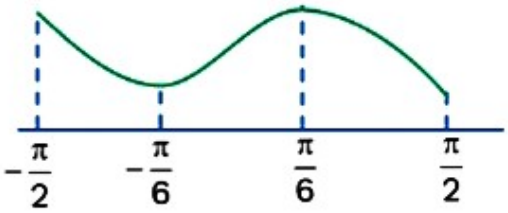
$\Rightarrow x = n\pi \pm \frac{\pi}{6}$

In $x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$, $f'(x) = 0$ at $x = \frac{\pi}{6}, -\frac{\pi}{6}$

$2\cos 2x - 1 > 0 \; \forall \; x \in \left(-\frac{\pi}{6}, \frac{\pi}{6}\right)$

$\Rightarrow f'(x) > 0 \; \forall \; x \in \left(-\frac{\pi}{6}, \frac{\pi}{6}\right)$

Also $f'(x) < 0 \; \forall \; x \in \left(-\frac{\pi}{2}, -\frac{\pi}{6}\right) \cup \left(\frac{\pi}{6}, \frac{\pi}{2}\right)$



So graph will be like:

$\Rightarrow$ So checking on critical points and end points, i.e., $\frac{-\pi}{2}, \frac{-\pi}{6}, \frac{\pi}{6}, \frac{\pi}{2}$

We get

$f\left(\frac{-\pi}{2}\right) = \frac{\pi}{2}$

$f\left(\frac{-\pi}{6}\right) = \frac{-\sqrt{3}}{2} + \frac{\pi}{6}$

$f\left(\frac{\pi}{6}\right) = \frac{\sqrt{3}}{2} - \frac{\pi}{6}$

$f\left(\frac{\pi}{2}\right) = -\frac{\pi}{2}$

$\Rightarrow f_{max} = \frac{\pi}{2}$ and $f_{min} = \frac{-\pi}{2}$

$\Rightarrow$ Difference between greatest and least value $= \frac{\pi}{2} - \left(\frac{-\pi}{2}\right) = \pi$.

Q.4 Discuss the extremum of $f(x) = 2x + 3x^{\frac{2}{3}}$

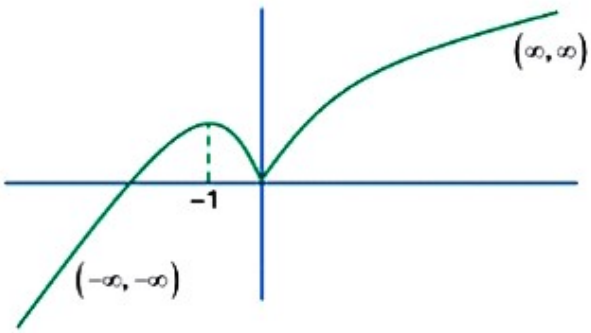
Sol. $f'(x) = 2 + \frac{2}{x^{\frac{1}{3}}} = \frac{2\left(x^{\frac{1}{3}} + 1\right)}{x^{\frac{1}{3}}}$



$\Rightarrow$ f(x) is not differentiable at x = 0

Also, f(x) is continuous.

So, graph will be :



$\therefore$ Range of graph is R.

Q.5 **Find maximum value of** $f(x) = x^2\sqrt{1-4x^2}$.

Sol. $1-4x^2 \geq 0$

$\Rightarrow x \in \left[\frac{-1}{2}, \frac{1}{2}\right]$

Let $x = \frac{1}{2}\sin\theta, \theta \in \left[\frac{-\pi}{2}, \frac{\pi}{2}\right]$

$\Rightarrow f(x) = y = \frac{\sin^2\theta\cos\theta}{4}$

Applying A.M $\geq$ G.M

$$\frac{\frac{\sin^2\theta}{2} + \frac{\sin^2\theta}{2} + \cos^2\theta}{3} \geq \left(\frac{\sin^4\theta\cos^2\theta}{4}\right)^{\frac{1}{3}}$$

$$\Rightarrow \left(\frac{1}{3}\right)^3 \geq \frac{\sin^4\theta\cos^2\theta}{4}$$

$$\Rightarrow \sin^4\theta\cos^2\theta \leq \frac{4}{27}$$

$$\Rightarrow \sin^2\theta\cos\theta \leq \frac{2}{\sqrt{27}} = \frac{2}{3\sqrt{3}}$$

$$\Rightarrow y = \frac{\sin^2\theta\cos\theta}{4} \leq \frac{1}{6\sqrt{3}}$$

$$\therefore f_{max} = \frac{1}{6\sqrt{3}}$$

Q.6 **Find maximum value of** $\sin\theta(1+\cos\theta)$.

Sol. $y = 2\sin\frac{\theta}{2}\cos\frac{\theta}{2}\left(2\cos^2\frac{\theta}{2}\right)$

$= 4\sin\frac{\theta}{2}\cos^3\frac{\theta}{2}$

Applying A.M. $\geq$ G.M.

$$\frac{\sin^2 x + \frac{\cos^2 x}{3} + \frac{\cos^2 x}{3} + \frac{\cos^2 x}{3}}{4} \geq \left(\frac{\sin^2 x\cos^6 x}{27}\right)^{\frac{1}{4}}$$

$$\Rightarrow \left(\frac{1}{4}\right)^4 \geq \frac{\sin^2 x \cos^6 x}{27}$$

$$\Rightarrow \frac{1}{16} \geq \frac{\sin x \cos^3 x}{3\sqrt{3}}$$

$$\Rightarrow \sin x \cos^3 x \leq \frac{3\sqrt{3}}{16}$$

$$\Rightarrow y_{max} = \frac{3\sqrt{3}}{4}$$

Q.7 **Find maximum value of** $\sin^3\theta\cos\theta$, $\theta \in \left(0, \frac{\pi}{2}\right)$.

Sol. $$\frac{\frac{\sin^2\theta}{3} + \frac{\sin^2\theta}{3} + \frac{\sin^2\theta}{3} + \cos^2\theta}{4} \geq \left(\frac{\sin^6\theta\cos^2\theta}{27}\right)^{\frac{1}{4}}$$

$$\Rightarrow \sin^3\theta\cos\theta \leq \frac{3\sqrt{3}}{16}$$ (∵ Previous Question).

PROBLEMS BASED ON MENSURATION AND GEOMETRY

Summary-Working Rule :

(1) When possible, draw a figure to illustrate the problem & label those parts that are important in the problem. Constants & variables should be clearly distinguished.

(2) Write an equation for the quantity that is to be maximised or minimised. This may require some algebraic manipulations.

(3) If y = f(x) is a quantity to be maximum or minimum, find those values of x for which $\frac{dy}{dx} = f'(x) = 0$.

(4) Test each values of x for which f'(x) = 0 to determine whether it provides a maximum or minimum or neither. The usual tests are:

(a) If $\frac{d^2y}{dx^2}$ is positive when $\frac{dy}{dx} = 0 \Rightarrow y$ is minimum.

If $\frac{d^2y}{dx^2}$ is negative when $\frac{dy}{dx} = 0 \Rightarrow y$ is maximum.

If $\frac{d^2y}{dx^2} = 0$ when $\frac{dy}{dx} = 0$, the test fails.

(b) If $\frac{dy}{dx}$ is $\left.\begin{array}{ll} \text{positive} & \text{for } x < x_0 \\ \text{zero} & \text{for } x = x_0 \\ \text{negative} & \text{for } x > x_0 \end{array}\right] \Rightarrow$ a maximum

occurs at $x = x_0$.

But if $\frac{dy}{dx}$ changes sign from negative to zero to positive as x advances through x_0 there is a minimum. If $\frac{dy}{dx}$ does not change sign, neither a maximum nor a minimum occurs. Such points are called INFLECTION POINTS.

(5) If the function $y = f(x)$ is defined for only a limited range of values $a \le x \le b$ then examine $x = a$ and $x = b$ for possible extreme values.

(6) If the derivative fails to exist at some point, examine this point as possible maximum or minimum.

Useful formulae of Mensuration to remember :

Volume of a cuboid = lbh.

Surface area of a cuboid = 2(lb + bh + hl).

Volume of a prism = areas of the base x height.

Lateral surface of a prism = perimeter of the base x height.

Total surface of a prism = lateral surface area + 2 area of the base (Note that lateral surfaces of a prism are all rectangles).

Volume of a pyramid

$= \frac{1}{3}$ (area of the base) × (height).

Curved surface area of a pyramid

$= \frac{1}{2}$ (perimeter of the base) × slant height.

(Note that slant surfaces of pyramid are triangles).

Volume of a cone $= \frac{1}{3}\pi r^2 h$.

Curved surface of a cylinder $= 2\pi rh$.

Total surface of a cylinder $= 2\pi rh + 2\pi r^2$.

Volume of a sphere $= \frac{4}{3}\pi r^3$.

Area of circular sector $= \frac{1}{2}r^2\theta$., when θ is in radians.

Q. **Find two positive numbers x and y such that x + y = 60 and x^3y is maximum.**

Sol. $x + y = 60$

$\Rightarrow y = 60 - x$

$\Rightarrow x^3y = (60 - x)x^3$

$f(x) = (60 - x)x^3; x \in (0, 60)$

For maximizing f(x), let us find critical points

$f'(x) = 3x^2(60 - x) - x^3 = 0$

$f'(x) = x^2(180 - 4x) = 0$

$\Rightarrow x = 45 \quad (\because x \neq 0)$

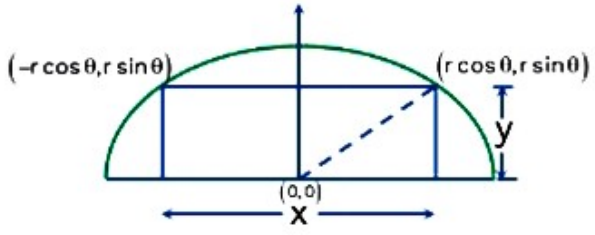
$f'(45^+) < 0$ and $f'(45^-) > 0$

Hence local maxima at x = 45.
So x = 45 and y = 15

Q. **Rectangles are inscribed inside a semi-circle of radius r. Find the rectangle with maximum area.**

Sol. Let us choose co-ordinate system with origin as centre of circle
Area, A = xy

$\Rightarrow A = 2(r\cos\theta)(r\sin\theta), \theta \in \left(0, \frac{\pi}{2}\right)$



$\Rightarrow A = r^2 \sin 2\theta$

$\Rightarrow \theta = \frac{\pi}{4}$ (for maximum area)

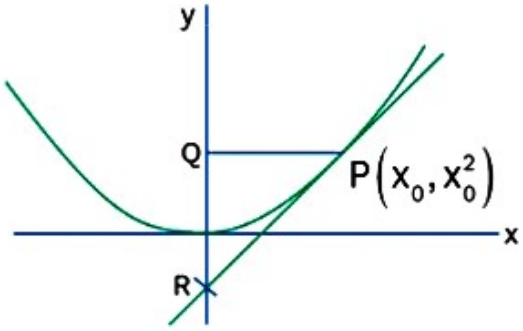
$\Rightarrow$ Sides of the rectangle are $2r\cos\left(\frac{\pi}{4}\right) = r\sqrt{2}$ and $r\sin\left(\frac{\pi}{4}\right) = \frac{r}{\sqrt{2}}$.

Q. **The tangent to the parabola $y = x^2$ has been drawn so that the abscissa x_0 of the point of tangency belong to the interval [1, 2]. Find x_0 for which the triangle to be bounded by the tangent, the axis of ordinates, and the straight line $y = x_0^2$ has the greatest area.**

Sol. $y = x^2, \frac{dy}{dx} = 2x$

$\Rightarrow$ Equation of the tangent at $\left(x_0, x_0^2\right)$ is

$$y - x_0^2 = 2x_0\left(x - x_0\right).$$



It meets y-axis at $R\left(0, -x_0^2\right)$

Q is $\left(0, x_0^2\right)$

$\Rightarrow$ Z = area of the triangle PQR

$$= \frac{1}{2} 2x_0^2 x_0 = x_0^3, 1 \le x_0 \le 2$$

$$\frac{dZ}{dx_0} = 3x_0^2 > 0 \text{ in } 1 \le x_0 \le 2$$

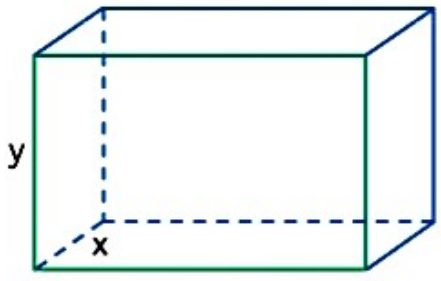
$\Rightarrow$ Z is an increasing function in [1, 2]

Hence, Z, i.e., the area of ΔPQR is greatest at $x_0 = 2$.

Q. **A sheet of area 40m² is used to make an open tank with square base. Find the dimensions of the base such that volume of this tank is maximum.**

Sol. Let the length of base be x m and height be y m

Volume $\tilde{O} = {}^2$



$\Rightarrow x^2 + 4xy = 40$

$$y = \frac{40 - x^2}{4x}, x \in \left(0, \sqrt{40}\right)$$

$$\Rightarrow V(x) = x^2\left(\frac{40 - x^2}{4x}\right) = \frac{40x - x^3}{4}$$

Maximizing volume,

$$V'(x) = \frac{40 - 3x^2}{4} = 0 \Rightarrow x = \sqrt{\frac{40}{3}} m$$

and $V''(x) = \frac{-3x}{2} \Rightarrow V''\left(\sqrt{\frac{40}{3}}\right) < 0$

$\Rightarrow$ Volume is maximum at $x = \sqrt{\frac{40}{3}} m$.

Q. **If a right-circular cylinder is inscribed in a given cone. Find the dimensions of the cylinder such that its volume is maximum.**

Sol. Let x be the radius of cylinder and y be its height

Volume $V = \pi x^2 y$

x, y can be related by using similar triangles

$$\frac{y}{r-x} = \frac{h}{r}$$

$$\Rightarrow y = \frac{h}{r}(r-x)$$

$$\Rightarrow V(x) = \pi x^2 \frac{h}{r}(r-x), x \in (0,r)$$

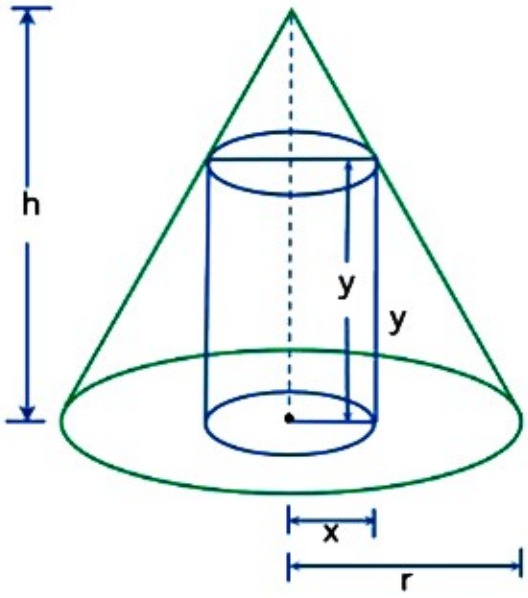
$$\Rightarrow V(x) = \frac{\pi h}{r}\left(rx^2 - x^3\right)$$

$$\Rightarrow V'(x) = \frac{\pi h}{r}\left(2rx - 3x^2\right)$$

$$V'(x) = 0 \Rightarrow x = \frac{2r}{3}$$

Also $V''(x) = \frac{\pi h}{r}(2r - 6x) \Rightarrow V''\left(\frac{2r}{3}\right) < 0$

Thus, volume is maximum when, $x = \frac{2r}{3}$ and $y = \frac{h}{3}$.



Q.1 **The vertical angle of a cone of maximum volume and given slant height is**

(A) $\tan^{-1}\sqrt{2}$ **(B)** $2\tan^{-1}\sqrt{2}$ **(C)** $\tan^{-1}\sqrt{3}$ **(D)** $2\tan^{-1}\sqrt{3}$

Sol. **(B)**

Given: ℓ is constant

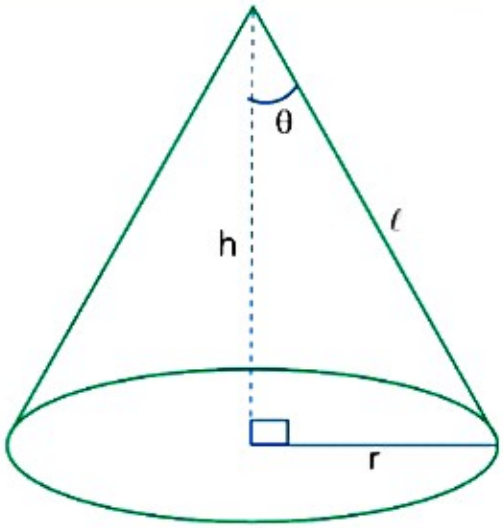
$h = \ell\cos\theta$, $r = \ell\sin\theta$

Volume $= V = \frac{1}{3}\pi r^2 h$

$$\Rightarrow V = \frac{1}{3}\pi\ell^3 \sin^2\theta\cos\theta$$

Applying AM $\geq$ GM

$$\frac{\frac{\sin^2\theta}{2} + \frac{\sin^2\theta}{2} + \cos^2\theta}{3} \geq \left(\frac{\sin^4\theta\cos^2\theta}{4}\right)^{\frac{1}{3}}$$



Equality holds when $\frac{\sin^2\theta}{2} = \cos^2\theta$

$\Rightarrow \tan^2\theta = 2$

$\Rightarrow \theta = \tan^{-1}\sqrt{2}$

$\therefore$ Vertical angle of cone of maximum volume $= 2\theta = 2\tan^{-1}\sqrt{2}$

Q.2 **A closed cylinder of given volume will have least surface area when the ratio of its height and base radius is**

(A) 2 : 1 **(B) 1 : 2** **(C) 2 : 3** **(D) 3 : 2**

Sol. **(A)**

$V = \pi r^2 h$

$S = 2\pi rh + 2\pi r^2$

$= 2\left(\frac{V}{r} + \pi r^2\right)$

$\Rightarrow S = 2\left(\frac{V}{2r} + \frac{V}{2r} + \pi r^2\right)$

$\Rightarrow$ Applying AM $\geq$ GM

$$\frac{\frac{V}{2r} + \frac{V}{2r} + \pi r^2}{3} \geq \left(\frac{V^2\pi}{4}\right)^{\frac{1}{3}}$$

Least value will occur when $\frac{V}{2r} = \pi r^2$

$\Rightarrow V = 2\pi r^3 = \pi r^2 h$

$\Rightarrow 2 = \frac{h}{r}$

Q.3 **Of all the closed cylindrical cans (right circular), which enclose a given volume of 10 cm³, which one has the minimum total surface area? Find the surface area.**

Sol. As seen in previous question, for least surface area, h = 2r

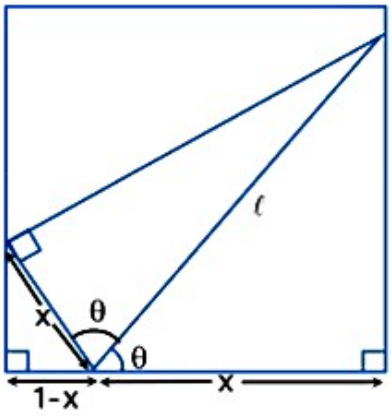
$\Rightarrow V = 2\pi r^3 = 10$

$\Rightarrow r = \left(\frac{5}{\pi}\right)^{\frac{1}{3}}$

$$S = 2\pi rh + 2\pi r^2$$
$$= 6\pi r^2$$
$$= 6\pi\left(\frac{5}{\pi}\right)^{\frac{2}{3}}$$
$$= 6\pi^{\frac{1}{3}}5^{\frac{2}{3}}\text{cm}^2.$$

Q.4 **One corner of a long rectangular sheet of paper of width 1 unit is folded over so as to reach the opposite edge of the sheet. Find the minimum length of the crease.**

Sol.



$x = \ell\cos\theta$

$x\cos(\pi - 2\theta) = 1 - x$

$\Rightarrow -x\cos 2\theta = 1 - x$

$$\Rightarrow x = \frac{1}{1-\cos 2\theta} = \frac{1}{2\sin^2\theta}$$

$$\Rightarrow \ell = \frac{1}{2\sin^2\theta\cos\theta}$$

$\therefore$ For ℓ_{min}, we have to find

maximum value of $\sin^2\theta\cos\theta$

$\Rightarrow \tan\theta = \sqrt{2}$ ($\because$ Question 1)

$$\Rightarrow \sin\theta = \frac{\sqrt{2}}{\sqrt{3}} \text{ and } \cos\theta = \frac{1}{\sqrt{3}}$$

$$\Rightarrow \ell_{min} = \frac{1}{2\left(\frac{2}{3}\right)\left(\frac{1}{\sqrt{3}}\right)} = \frac{3\sqrt{3}}{4}$$

Q.5 **A rectangular sheet of fixed parameter having lengths in ratio 8:15 is converted into an open rectangular box by folding after removing squares of equal area for all four corners. If total area of removed squares is 100, the resulting box has maximum volume. Then the lengths of sides of rectangular sheet are**

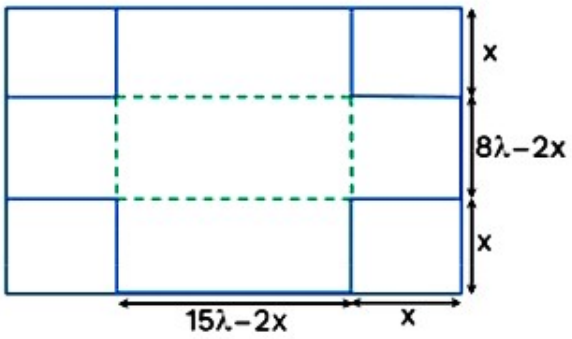
Sol. Area of removed squares $= 4x^2 = 100$

$\Rightarrow x = 5$

Volume of open box

$$= x(8\lambda - 2x)(15\lambda - 2x)$$

$$\Rightarrow V(x) = 4x^3 - 46\lambda x^2 + 120\lambda^2 x$$



For x = 5, volume is maximum.

$\Rightarrow V'(x) = 0$ for $x = 5$

$\Rightarrow 12x^2 - 92\lambda x + 120\lambda^2 = 0$ for $x = 5$

$\Rightarrow 3(25) - 23(5)\lambda + 30\lambda^2 = 0$

$\Rightarrow \lambda = 3$ or $\frac{5}{6}$

$V''(x) = 24x - 92\lambda$

At $x = 5$, $V''(x) = 120 - 92\lambda$

$\therefore$, for $\lambda = 3, V(x)$ is maximum while it is minimum for $\lambda = \frac{5}{6}$

$\left(\because \frac{d^2y}{dx^2} < 0 \text{ at maxima}\right)$

So length of sides of rectangles are 24 and 45.

Q.6 **For a train/steamer the cost of fuel per hour varies as the square of its speed (in km/hr) and the cost is ₹ 24/hr when the speed is 12 km/hr. If other expenses amount to ₹ 96/hr, find the most economical speed and the cost of journey for 100 km.**

Sol. Let fuel cost be $F = \lambda v^2$ where v is velocity

$\Rightarrow 24 = \lambda(12)^2$

$\Rightarrow \lambda = \frac{1}{6}$

$\Rightarrow F = \frac{v^2}{6}$

$$\text{Total cost} = 96\left(\frac{100}{v}\right) + \frac{v^2}{6}\left(\frac{100}{v}\right)$$

$$= \left(\frac{9600}{v} + \frac{50v}{3}\right)$$

Applying AM $\geq$ GM,

$$\frac{\frac{9600}{v}+\frac{50v}{3}}{2} \geq 10\sqrt{\left(\frac{96}{v}\right)\left(\frac{50v}{3}\right)}$$

$$\Rightarrow \frac{9600}{v}+\frac{50v}{3} \geq 800$$

$\therefore$ Minimum cost = ₹ 800

It occurs when $\frac{9600}{v}=\frac{50v}{3}$

$\Rightarrow v^2 = 576$

$\Rightarrow v = 24\text{km / hr}$ (Most economical speed)

Q.7 **Find the coordinates of all the points P on the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ for which the area of the Δ PON is maximum, where O denotes the origin and N, the foot of the perpendicular from O to the tangent at P. (a > b)**

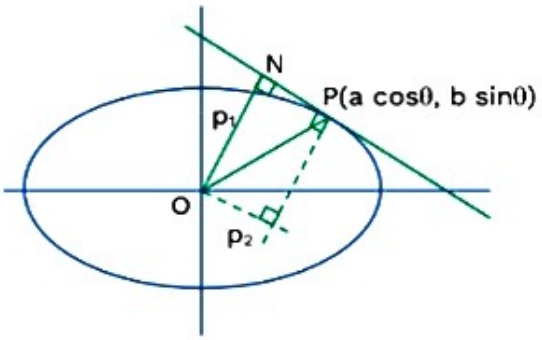
Sol. Area of $\Delta PON = \frac{1}{2}p_1 p_2$

Equation of tangent is $\frac{x\cos\theta}{a}+\frac{y\sin\theta}{b}=1$

Equation of normal at P is $ax\sec\theta - by\,\text{cosec}\theta = a^2 - b^2$

$$p_1 = \frac{1}{\sqrt{\frac{\cos^2\theta}{a^2}+\frac{\sin^2\theta}{b^2}}}$$

$$p_2 = \frac{a^2-b^2}{\sqrt{a^2\sec^2\theta + b^2\,\text{cosec}^2\theta}}$$



$$\text{Area} = \frac{1}{2}\frac{\left(a^2-b^2\right)}{\sqrt{2+\frac{b^2}{a^2}\cot^2\theta+\frac{a^2}{b^2}\tan^2\theta}}$$

Since $\frac{b^2\cot^2\theta}{a^2}+\frac{a^2}{b^2}\tan^2\theta \geq 2 \quad \left(\because t+\frac{1}{t}\geq 2\right)$

$$\text{Maximum Area} = \frac{1}{2}\frac{\left(a^2-b^2\right)}{\sqrt{2+2}}$$

$$= \frac{a^2-b^2}{4} \text{ when } \tan^2\theta = \frac{b^2}{a^2}$$

$\Rightarrow \tan\theta = \frac{b}{a}, \frac{-b}{a}$

Hence there are 4 possible coordinates of P, one in each quadrant.

$P \equiv \left(\pm \frac{a^2}{\sqrt{a^2+b^2}}, \pm \frac{b^2}{\sqrt{a^2+b^2}} \right).$

Q.8 **A wire of length 20 cm is cut in 2 pieces; one piece converted in circle and other into square. Where the wire should be cut from so that the sum of total area of 2 plane figure is minimum; maximum?**

Sol. $x = 2\pi r \quad \Rightarrow r = \frac{x}{2\pi}$

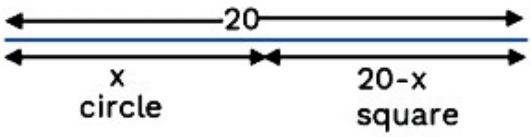
$20 - x = 4a \Rightarrow a = \frac{20-x}{4}$

Area of circle $= \frac{x^2}{4\pi}$

Area of square $= \left(\frac{20-x}{4} \right)^2$

Total area $= \frac{x^2}{4\pi} + \frac{(20-x)^2}{16} = A(x)$

$A'(x) = \frac{2x}{4\pi} - \frac{(20-x)}{8} > 0$



$\Rightarrow 4x > 20\pi - \pi x$

$\Rightarrow x > \frac{20\pi}{\pi+4}$

So, A(x) is minimum at $x = \frac{20\pi}{\pi+4}$

So, maximum area will occur at end points

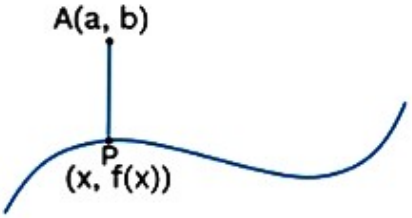
$A(0) = 25$

$A(20) = \frac{100}{\pi}$

So, maximum area is $\frac{100}{\pi}$

GENERAL CONCEPT :

Given a fixed point A(a, b) and a moving point P(x,f(x)) on the curve y=f(x). Then AP will be maximum or minimum if it is normal to the curve at P.



Significance of the 2nd order Derivative and Point of Inflection:

A point where the graph of function is continuous and has a tangent line and where the concavity changes is called point of inflection.

At the point of inflection either y" = 0 and changes sign or y" fails to exist.

At the point of inflection, the curve crosses its tangents.

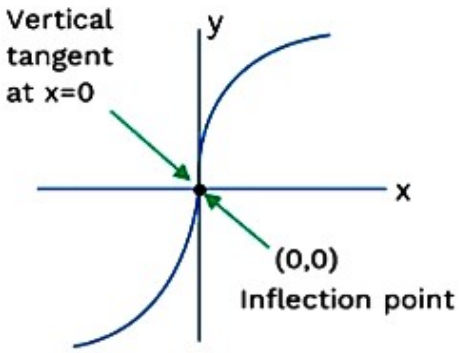
A function can not have point of inflection and extrema at same point.

Note :

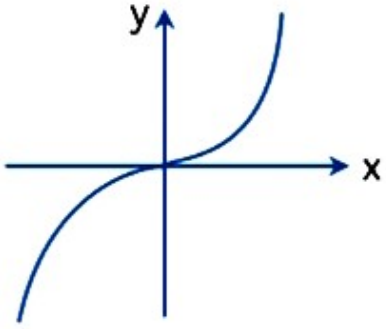
If $\frac{d^2y}{dx^2} > 0$ then y is concave up and if $\frac{d^2y}{dx^2} < 0$ then y is concave down.

Ex. $f(x) = x^{\frac{1}{5}}$ at x = 0 has inflection point

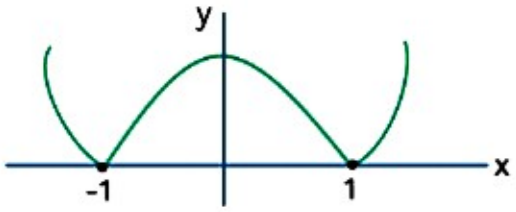
y" D.N.E. at x = 0
Note that f(x) has a vertical tangent and the curve crosses its tangent at x = 0



Ex. $f(x) = x^3$ at x = 0 has inflection point.
y" = 0 at x = 0 and changes sign at x = 0



Ex. $f(x) = |x^2 - 1|$ has no inflection point in its domain. $x = 1$ or -1 are not the points of inflection as no tangent can be drawn at these points, even if concavity changes at these points.



Q. **Number of points of inflection for $f(x) = x^2e^{-|x|}$ is**

(A) 1 **(B) 2** **(C) 3** **(D) 4**

Sol. We have $f(x) = x^2e^{-|x|} = \begin{cases} x^2e^{-x}, & x \geq 0 \\ x^2e^{x}, & x < 0 \end{cases}$

$$\Rightarrow f'(x) = \begin{cases} e^{-x}(2x - x^2), & x \geq 0 \\ e^{x}(x^2 + 2x), & x < 0 \end{cases}$$

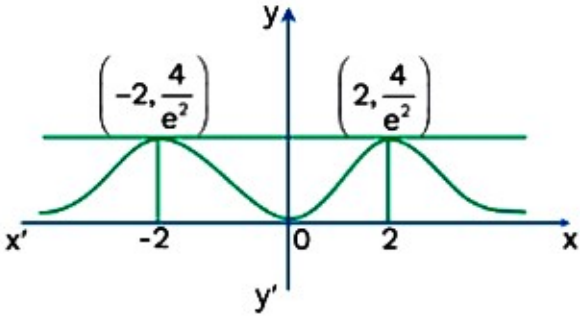
$f(x)$ increases in $(-\infty, -2) \cup (0, 2)$

and $f(x)$ decreases in $(-2, 0) \cup (2, \infty)$

$$\Rightarrow f''(x) = \begin{cases} e^{-x}(x^2 - 4x + 2), & x \geq 0 \\ e^{x}(x^2 + 4x + 2), & x < 0 \end{cases}$$

$f''(x) = 0$ at $-2 \pm \sqrt{2}$ and $2 \pm \sqrt{2}$.

So, there are 4 points of inflection.



Q.1 **Check for point of inflection for $f(x) = x^4$ at $x = 0$.**

Sol. $f'(x) = 4x^3$

$f''(x) = 12x^2$

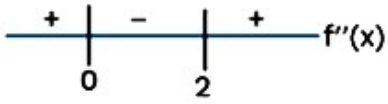
Here, while both $f'(0)$ and $f''(0) = 0$, $f''(x)$ does not change the sign at $x = 0$.
$\therefore$, There will be no point of inflection at $x = 0$.

Q.2 **Sketch $y = f(x) = x^4 - 4x^3 + 10$**

Sol. $f'(x) = 4x^3 - 12x^2$
$= 4x^2 (x - 3)$



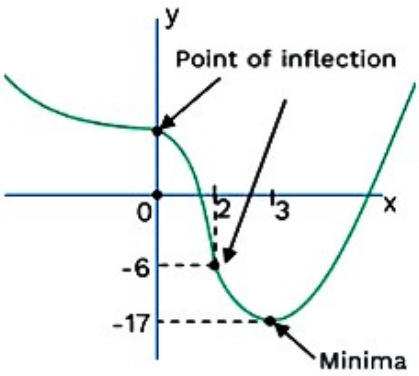
$f''(x) = 12x^2 - 24x$
$= 12x (x - 2)$



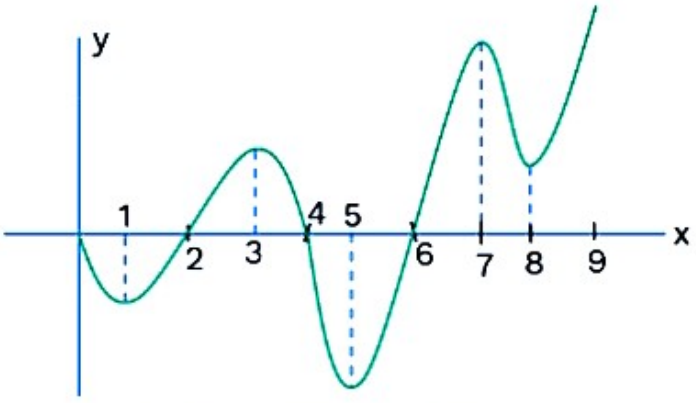
So graph will be concave up for $x \in (-\infty, 0) \cup (2, \infty)$.

Graph will be concave down for $x \in (0, 2)$.

Graph will be decreasing for $x < 3$ and after that, it will be increasing.
Also there will be point of inflection at $x = 0$ and $x = 2$.
$f(-\infty) = \infty$;
$f(\infty) = \infty$;
$f(3) = -17$;
$f(0) = 10$;
$f(2) = -6$
Now plot these points



Q.3 **The graph of the first derivative f' of a function is shown.**



(a) On what intervals is f increasing? Explain.
(b) At what values of x does f have local minima or maxima?
(c) On what intervals is f concave upwards or concave downwards?
(d) What are x-coordinates of the inflection points of f?

Sol. **(a)** f is increasing when $f'(x) > 0$.

$\therefore$ f is increasing in $(2,4) \cup (6,\infty)$

(b) At local minima, $f'(x) = 0$ and it changes from negative to positive value.
So, f has minima at $x = 2$ and $x = 6$.
At local maxima, $f'(x) = 0$ and it changes from positive to negative value.
So, f has maxima at $x = 4$.

(c) f is concave upwards when $f'' > 0$, i.e, f' is increasing while f is concave downwards when $f'' < 0$.

$\therefore$ f is concave upwards in $(1,3) \cup (5,7) \cup (8,\infty)$

f is concave downwards in $(0,1) \cup (3,5) \cup (7,8)$

(d) At point of inflection, $f''(x) = 0$ and it changes sign,
So, check at point extrema of $f'(x)$.
$\Rightarrow x = 1, 3, 5, 7, 8$ are points of inflection.

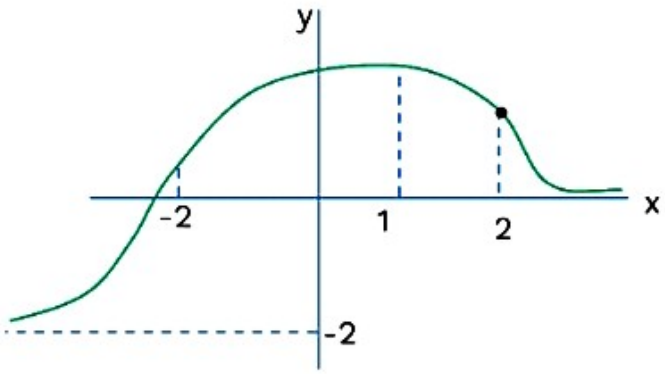
Q.4 **Sketch a possible graph of a differentiable function f that satisfies the following condition :**

(i) $f'(x) > 0$ on $(-\infty,1), f'(x) < 0$ on $(1,\infty)$

(ii) $f''(x) > 0$ on $(-\infty,-2) \cup (2,\infty), f''(x) < 0$ on $(-2,2)$

(iii) $\lim\limits_{x\to-\infty} f(x) = -2, \lim\limits_{x\to\infty} f(x) = 0$

Sol.

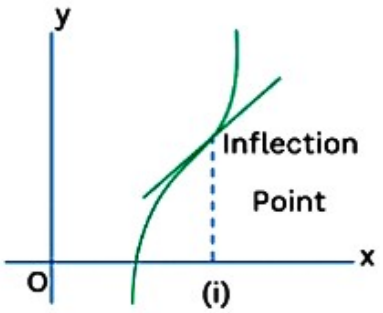


Graph is concave down in (–2, 2) and concave up everywhere else.

Different Graphs of the cubic :

$y = ax^3 + bx^2 + cx + d$

(1) One real & two imaginary roots.
(always monotonic) $\forall x \in R$



(i)

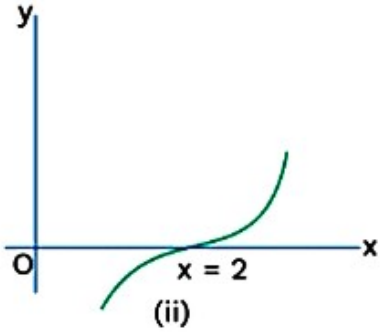
Condition :

$f'(x) \geq 0$ of $f'(x) \leq 0$ together with either $f'(x) = 0$ has no root (i.e. $D<0$) or $f'(x) = 0$ has a root $x = \alpha$ then $f(\alpha) = 0$.

(i) either $f'(x) = 0$ has no real root

Or **(ii)** if $f'(x) = 0$ has a root $x = \alpha$ then

$f(\alpha) = 0$



(ii)

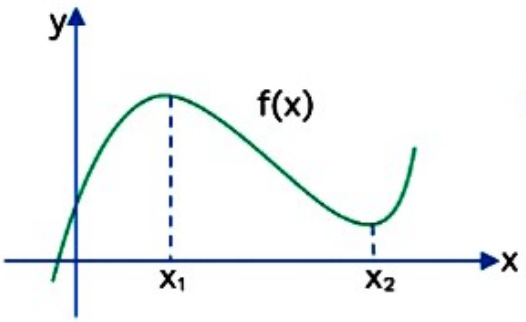
e.g. $y = x^3 - 2x^2 + 5x + 4$
$y' = 3x^2 - 4x + 5$ $(D < 0)$

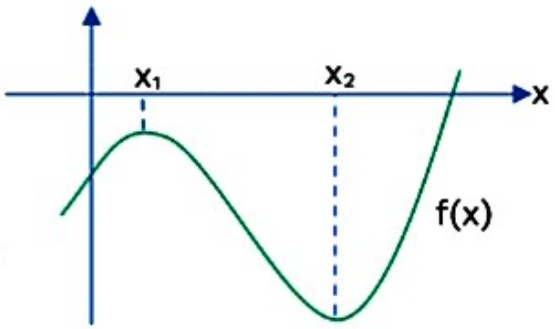
e.g. $y = (x - 2)^3$
$y' = 3(x - 2)^2 = 0 \Rightarrow x = 2$
also $f(2) = 0$
gives $x = 2$, $y(2) = 0$

Note :

In this case if $f'(x) = 0$ has a root $x = \alpha$ and $f(\alpha) = 0$ this would mean $f(x) = 0$ has repeated roots which is dealt separately.

(2) Exactly one root and non monotonic.

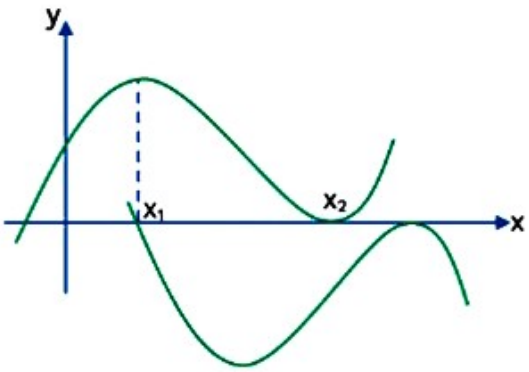




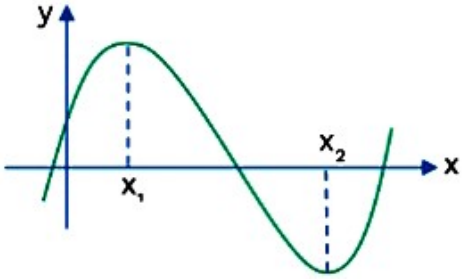
$f(x_1).f(x_2) > 0$
Where x_1 and x_2 are the roots of $f'(x) = 0$

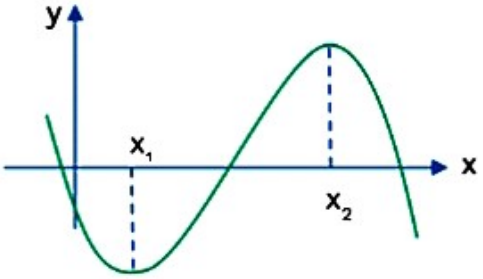
(3) Three roots $\begin{cases} \text{Two coincident} \\ \text{One different} \end{cases}$

$f(x_1).f(x_2) = 0$



(4) All three distinct real roots



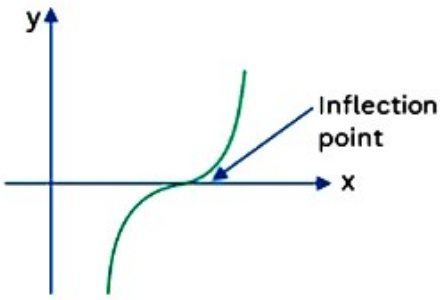


$f(x_1).f(x_2) < 0$
Where x_1 and x_2 are the roots of $f'(x) = 0$

(5) All three roots coincident

$f'(x) \geq 0$ or $f'(x) \leq 0$ and $f(\alpha) = 0$

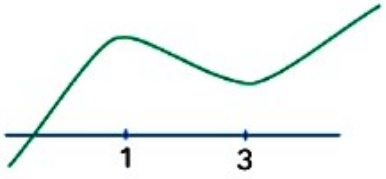
where α is a root of $f'(x) = 0$
e.g. $y = (x - 1)^3$



Note :
Graph of every cubic polynomial must have one point of inflection.

Q.1 **Find the value of a if $x^3 - 3x + a = 0$ has three real distinct roots.**

Sol. Let $f(x) = x^3 - 3x + a$
Let $f'(x) = 0$
$\Rightarrow 3x^2 - 3 = 0 \Rightarrow x = \pm 1$
For three distinct roots, $f(1)f(-1) < 0$
$\Rightarrow (1 - 3 + a)(-1 + 3 + a) < 0$
$\Rightarrow (a + 2)(a - 2) < 0$
$\Rightarrow -2 < a < 2$



Q. **Prove that three exist exactly two non-similar isosceles triangle ABC such that tanA + tanB + tanC = 100.**

Sol. Let A = B, then 2A + C = 180° and 2tanA + tanC = 100

Now $2A + C = 180° \Rightarrow \tan 2A = -\tan C$ ...(i)

Also $2\tan A + \tan C = 100$

$\Rightarrow 2\tan A - 100 = -\tan C$ ...(ii)

From (i) and (ii), $2\tan A - 100 = \dfrac{2\tan A}{1-\tan^2 A}$

Let tanA = x, then $\dfrac{2x}{1-x^2} = 2x - 100$

$\Rightarrow x^3 - 50x^2 + 50 = 0$

Let $f(x) = x^3 - 50x^2 + 50$. Then $f'(x) = 3x^2 - 100x$. Thus $f'(x) = 0$ has roots $0, \dfrac{100}{3}$.

Also $f(0)f\left(\dfrac{100}{3}\right) < 0$. Thus f(x) = 0 has exactly three distinct real roots. Therefore, tanA and hence A has three distinct values but one of them will be obtuse angle. Hence, there exist exactly two non similar isosceles triangles.

Q. **Find the set of value of m so that the cubic $x^3 - \dfrac{3}{2}x^2 + \dfrac{5}{2} = \log_{\frac{1}{4}}(m)$ has 3 distinct solutions.**

Sol. Consider $y = x^3 - \dfrac{3}{2}x^2 + \dfrac{5}{2}$

$\dfrac{dy}{dx} = 3x^2 - 3x = 3x(x-1) = 0 \Rightarrow x = 0$ or 1

$\dfrac{d^2y}{dx^2} = 6x - 3; \left.\dfrac{d^2y}{dx^2}\right|_{x=0} = -3$ i.e. < 0

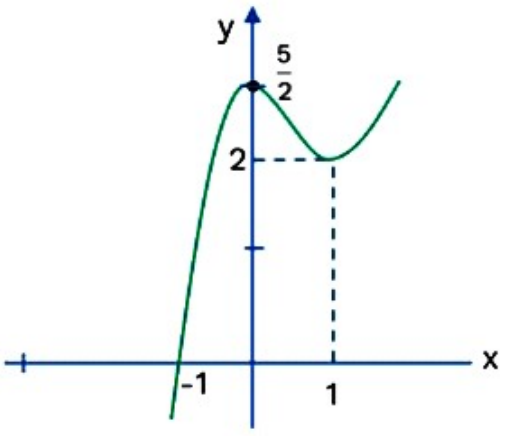
$\Rightarrow$ maxima at x = 0

$\left.\dfrac{d^2y}{dx^2}\right|_{x=1}$ = is 3 i.e. $> 0 \Rightarrow$ minima

Hence the graph of the cubic is given.

$\Rightarrow$ For 3 distinct roots

$2 < \log_{\frac{1}{4}}(m) < \dfrac{5}{2}$



$$2 < -\log_4(m) < \frac{5}{2}$$

$$-\frac{5}{2} < \log_4(m) < -2$$

$$\frac{1}{32} < m < \frac{1}{16}$$

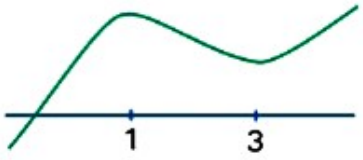
Q.1 $\frac{x^3}{3} - 2x^2 + 3x + 4 = 0$ **has how many number of real roots?**

Sol. $f(x) = \frac{x^3}{3} - 2x^2 + 3x + 4$

$\Rightarrow f'(x) = x^2 - 4x + 3 = (x-1)(x-3)$

$\Rightarrow f(1) = \frac{1}{3} - 2 + 3 + 4 = \frac{16}{3} > 0$

$\Rightarrow f(3) = 9 - 2\times 9 + 3\times 3 + 4 = 4 > 0$



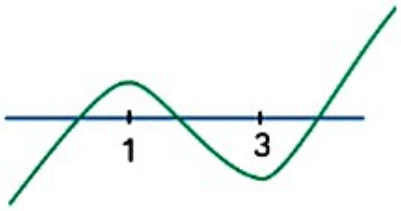
From graph, it can be seen that there is only 1 real root.

Q.2 $\frac{x^3}{3} - 2x^2 + 3x - 1 = 0$ **has how many number of real roots?**

Sol. $f'(x) = (x-1)(x-3)$

$f(1) = \frac{1}{3} > 0$ and $f(3) = -1 < 0$

Unlike previous question, f(3) is negative
From graph, it is clear that f(x) has 3 real roots.



Q.3 **If** $f(x) = x^3 - 3x + a, a \in (0,2), f(x) = 0$ **has 3 distinct roots** x_1, x_2, x_3**. Then find** $\{x_1\} + \{x_2\} + \{x_3\}$ **where {.} represents fractional part function.**

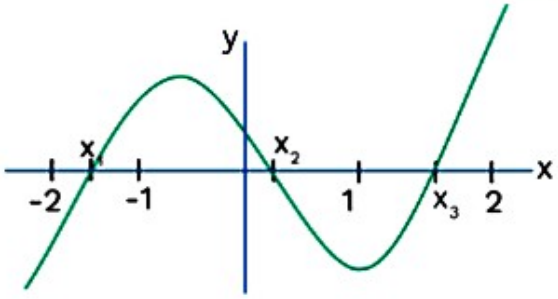
Sol. $f'(x) = 3x^2 - 3 = 3(x-1)(x+1)$

$f(1) = a - 2 < 0$

$f(-1) = a + 2 > 0$

$f(-2) = a - 2 < 0$

$f(0) = a > 0$



$f(2) = a + 2 > 0$

$\Rightarrow$ Graph:

$$\{x_1\} + \{x_2\} + \{x_3\} = x_1 - [x_1] + x_2 - [x_2] + x_3 - [x_3]$$

$$= x_1 + x_2 + x_3 - ([x_1] + [x_2] + [x_3])$$

$$= 0 - (-2 + 0 + 1)$$

$$= 1$$

Q.4 **P(x) is a polynomial of degree 3. P(–1)=10 and P(1) =–6. If P(x) has maxima at x = –1 and P′(x) has minima at x = 1, find P(x).**

Sol. Let $P''(x) = 6\lambda(x - 1)$ $\quad (\because P''(x) = 0 \text{ at } x = 1)$

$\Rightarrow P'(x) = 3\lambda(x - 1)^2 + c$

$P'(-1) = 0$ $\quad$ (P(x) has maxima at x = –1)

$\Rightarrow 12\lambda + c = 0$

$\Rightarrow c = -12\lambda$

$\Rightarrow P'(x) = 3\lambda(x - 1)^2 - 12\lambda$

$\Rightarrow P(x) = \lambda(x - 1)^3 - 12\lambda x + \mu$

Now, $P(-1) = 10$ and $P(1) = -6$

$\Rightarrow -8\lambda + 12\lambda + \mu = 10$ $\quad (\because P(-1) = 10)$

$\Rightarrow 4\lambda + \mu = 10$

Also, $-12\lambda + \mu = -6$ $\quad (\because, P(1) = -6)$

$\Rightarrow \mu = 6$ and $\lambda = 1$

$\Rightarrow P(x) = (x - 1)^3 - 12x + 6$

$\Rightarrow P(x) = x^3 - 3x^2 - 9x + 5$

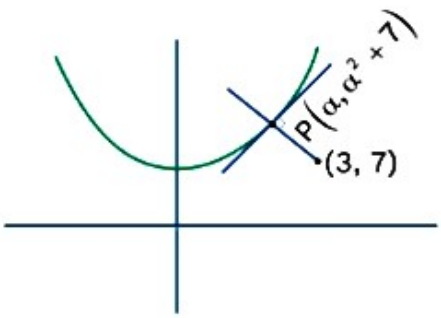
PROBLEMS ON SHORTEST DISTANCE BETWEEN TWO CURVES

Note :

Shortest distance is along the common normal.

Q.1 **A helicopter of enemy is flying along the curve given by $y = x^2 + 7$. A soldier placed at (3, 7) wants to shoot down the helicopter when it is nearest to him. Find the nearest distance.**

Sol. Let normal from (3, 7) cuts the curve at $P(\alpha, \alpha^2 + 7)$.



$$\text{Slope} = \frac{-1}{\frac{dy}{dx}} = \frac{-1}{2\alpha} = \frac{\alpha^2 + 7 - 7}{\alpha - 3}$$

$$\Rightarrow \frac{\alpha^2}{\alpha - 3} = \frac{-1}{2\alpha}$$

$$\Rightarrow 2\alpha^3 + \alpha - 3 = 0$$

$$\Rightarrow \alpha = 1 \Rightarrow P : (1, 8)$$

$$\therefore \text{Distance} = \sqrt{(3-1)^2 + (7-8)^2} = \sqrt{5}$$

Q.2 **A(−2, 3), B(−1, 1), C and D(2, 7) form a quadrilateral. If all point lie on $y = ax^2 + bx + c$, find C such that area of quadrilateral is maximum.**

Sol. Since A, B and D lie on $y = ax^2 + bx + c$,

$3 = 4a - 2b + c$

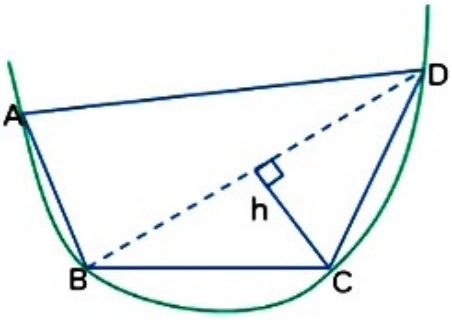
$1 = a - b + c$

$7 = 4a + 2b + c$

Solving it, we get $a = b = c = 1$

$$\Rightarrow y = x^2 + x + 1$$

$$\Rightarrow C(\alpha, \alpha^2 + \alpha + 1)$$



Area of quad. ABCD = Area of ΔABD + Area of ΔBCD

Since area of ΔABD is fixed, for maximum area, we should maximize the area of ΔBCD

Area of $\Delta BCD = \frac{1}{2}(BD)(h)$.

So, to maximize area, we have to maximize 'h'
For it tangent at C should be parallel to BD.

$\Rightarrow m(BD) = \frac{7-1}{2+1} = 2 = \frac{dy}{dx} = 2\alpha + 1$

$\Rightarrow \alpha = \frac{1}{2} \Rightarrow C : \left(\frac{1}{2}, \frac{7}{4}\right)$

Q.3 **Find point on ellipse $x^2 + 2y^2 = 6$ which is at minimum distance from line $x + y = 7$.**

Sol. Ellipse : $\frac{x^2}{6} + \frac{y^2}{3} = 1$

PQ will be shortest if it is common normal

Tangents through P is $\frac{xx_1}{6} + \frac{yy_1}{3} = 1$

Slope of tangent $= \frac{-x_1}{2y_1}$ = slope of line $x + y = 7$

$\Rightarrow \frac{-x_1}{2y_1} = -1$

$\Rightarrow x_1 = 2y_1$

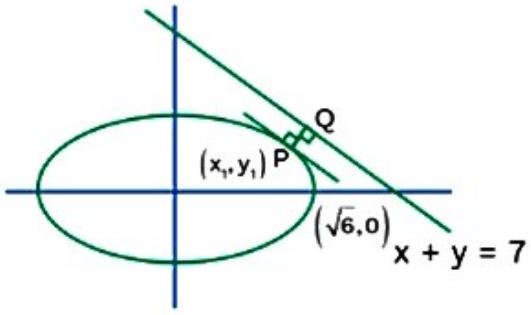
$\Rightarrow 4y_1^2 + 2y_1^2 = 6$

$\Rightarrow y_1 = 1$

($\because y_1 = -1$ is rejected from graph)

$\Rightarrow x_1 = 2$

$\Rightarrow P(2,1)$



Q.4 **The minimum value of** $\sqrt{(U-V)^2 + \left(\sqrt{2-U^2} - \frac{9}{V}\right)^2}$ **for** $0 < U < \sqrt{2}$ and $V > 0$ **is**

(A) $\sqrt{2}$ **(B)** $3\sqrt{2}$ **(C)** $4\sqrt{2}$ **(D)** $2\sqrt{2}$

Sol. Using distance formula, above expression actually asks for minimum distance between $x^2 + y^2 = 2$ and $xy = 9$.
So, line should be along common normal which also passes through center of circle.

Distance $OP = \sqrt{\alpha^2 + \frac{81}{\alpha^2}}$

$\Rightarrow OP_{min} = \sqrt{18}$

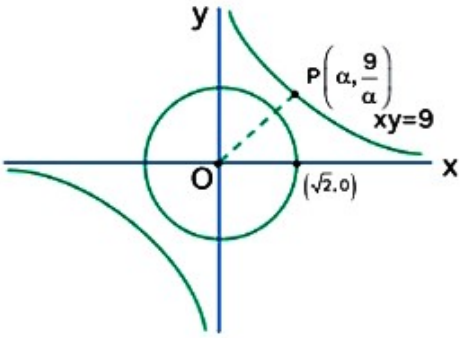
(using $AM \geq GM$ on α^2 and $81/\alpha^2$)

$\Rightarrow \alpha =$

Minimum distance $= OP - \text{radius}$

$= 3\sqrt{2} - \sqrt{2}$

$= 2\sqrt{2}$



Q.5 **Find shortest distance between curves $y = e^x$ and $y = \ln x$.**

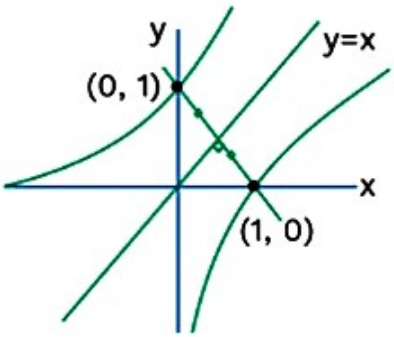
Sol. Both curves are inverse of each other and mirror image of each other about $y = x$.
By symmetry, we can see that shortest distance will be twice the shortest distance between curve and $y = x$
Since slope of $y = x$ is 1, slope of tangent at point of contact on curve is also 1

$\Rightarrow \frac{dy}{dx} = 1 \Rightarrow e^x = 1$

$\Rightarrow x = 0$

$\Rightarrow$ Point is (0, 1)

$\Rightarrow$ Distance between curves $= 2\left(\frac{1}{\sqrt{2}}\right) = \sqrt{2}$.



Q.6 **Find shortest distance between $y = x^2 + 3$ and $x = y^2 + 3$**

Sol. Again both curves are inverse of each other
Common normal is perpendicular to $y = x$.

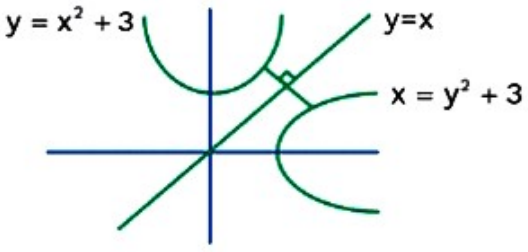
So, $\frac{dy}{dx} = 1$ (of tangent at point of contact)

$\Rightarrow 2x = 1$

$\Rightarrow x = \frac{1}{2}$ $\Rightarrow y = \frac{13}{4}$

It's mirror image will be $\left(\frac{13}{4}, \frac{1}{2}\right)$.

Distance $= \sqrt{\left(\frac{13}{4} - \frac{1}{2}\right)^2} \times 2 = \frac{11}{2\sqrt{2}}$



Q.1 **Find maximum and minimum distance of $x^2 + xy + y^2 = 1$ from (0, 0).**

Sol. Let $x = r\cos\theta$ and $y = r\sin\theta$

$\Rightarrow r^2\cos^2\theta + r^2\cos\theta\sin\theta + r^2\sin^2\theta = 1$

$\Rightarrow r^2 = \dfrac{1}{1+\sin\theta\cos\theta}$

$\Rightarrow r^2 = \dfrac{1}{1+\dfrac{\sin 2\theta}{2}}$

Since $\sin 2\theta \in [-1,1]$,

$\Rightarrow r^2 \in \left[\dfrac{2}{3}, 2\right]$

$\Rightarrow |r| \in \left[\sqrt{\dfrac{2}{3}}, \sqrt{2}\right]$

Distance of point from origin $= \sqrt{(r\cos\theta)^2 + (r\sin\theta)^2} = |r|$

$\therefore$ Minimum distance is $\sqrt{\dfrac{2}{3}}$.

Maximum distance is $\sqrt{2}$

Q.2 **Find the maximum distance of a point on $5x^2 - 8xy + 5y^2 = 4$ from origin.**

Sol. Let $x = r\cos\theta; y = r\sin\theta$

Distance from origin $= \sqrt{x^2 + y^2} = |r|$

Also, $5r^2\cos^2\theta - 8r^2\sin\theta\cos\theta + 5r^2\sin^2\theta = 4$

$\Rightarrow r^2 = \dfrac{4}{5 - 8\sin\theta\cos\theta}$

$\Rightarrow r^2 = \dfrac{4}{5 - 4\sin 2\theta}$

$\because \sin 2\theta \in [-1,1]$

$\Rightarrow r^2 \in \left[\dfrac{4}{9}, 4\right]$

$\Rightarrow |r| \in \left[\dfrac{2}{3}, 2\right]$ $\therefore$ Minimum distance $= \dfrac{2}{3}$

Solved Example

Q.1 **Find the shortest distance between the curves $y^2 = x^3$ and $9x^2 + 9y^2 - 30y + 16 = 0$**

Sol. We have $9x^2 + 9y^2 - 30y + 16 = 0$
Which is a circle having

$$\text{centre} \equiv \left(0, \frac{5}{3}\right) \text{and radius} = \sqrt{\left(\frac{5}{3}\right)^2 - \frac{16}{9}} = 1.$$

Let us choose any point on the curve $y^2 = x^3$ as $A\left(t^2, t^3\right)$. If B is the point on the circle and nearest to A(see figure), then

AB = AC − radius

$$= \sqrt{\left(t^2 - 0\right)^2 + \left(t^3 - \frac{5}{3}\right)^2} - 1$$

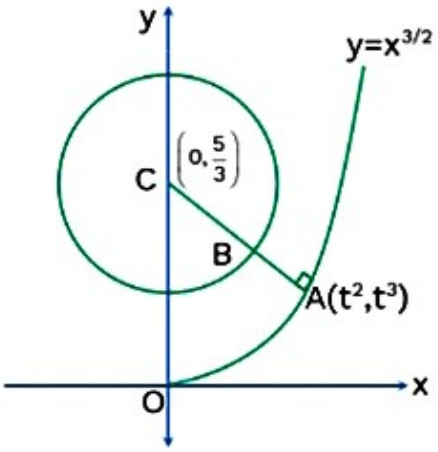
Let $f(t) = t^4 + \left(t^3 - \frac{5}{3}\right)^2$

And $f'(t) = 4t^3 + 2\left(t^3 - \frac{5}{3}\right).3t^2$

$$= 2t^2\left[2t + 3\left(t^3 - \frac{5}{3}\right)\right]$$

$$= 2t^2\left(3t^3 + 2t - 5\right)$$

$$= 2t^2(t-1)\left(3t^2 + 3t + 5\right)$$



The value of 't' at which AB attains minima is given by the equation

f'(t) = 0 which gives t = 0, 1

But t = 0 is not a point of extrema, as f'(t) does not change sign in the neighbourhood of t = 0.

However, t = 1 is a point of minima, since $f(1^-) < 0$ and $f'(1^+) > 0$.

Hence, the minimum value of AB is

$$\sqrt{1 + \left(1 - \frac{5}{3}\right)^2} - 1 \simeq 0.2$$

Q.2 **If the function $f(x)=(a+3)x^3+(a-3)x^2+4(a-4)x+5$ has maxima at some $x\in R^-$ and a minima at some $x\in R^+$, find the possible values of a.**

Sol. We have

$f(x)=(a+3)x^3+(a-3)x^2+4(a-4)x+5$

And $f'(x)=3(a+3)x^2+2(a-3)x+4(a-4)$

According to the given condition, $f'(x)$ must vanish at two real and distinct points say α,β such that $\alpha<0$ and $\beta>0$. Thus, we have

$f'(x)=3(a+3)(x-\alpha)(x-\beta)$

And $f''(x)=3(a+3)(x-\alpha+x-\beta)$.

According to the given condition, α is to be a maxima

i.e. $f''(\alpha)<0$

i.e. $3(a+3)(\alpha-\beta)<0$

i.e. $a+3>0 \qquad [\alpha-\beta<0]$

i.e. $a>-3$

and β is to be minima

i.e. $f''(\beta)>0$

i.e. $3(a+3)(\beta-\alpha)>0$

i.e. $a+3>0 \qquad [\beta-\alpha>0]$

i.e. $a>-3$

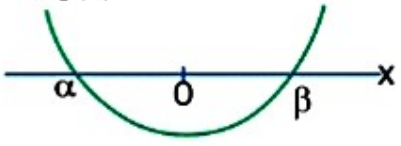
Thus, taking together all the above results $(a+3>0)$, the graph of $y=3(a+3)x^2+2(a-3)x+4(a-4)$ must look as shown below. For the curve to look like this, the necessary and sufficient condition is $y(0)<0$

i.e. $4(a-4)$

i.e. $a<4$

Hence, the possible values of a , are

$a\in(-3,4)$



$[a+3>0 \Rightarrow$ concavity upwards along the + ve X – Y axis$]$

Q.3 **Find a point M on the curve $y = \frac{3}{\sqrt{2}} x \ln x, x \in \left(e^{-1.5}, \infty\right)$ such that segment of the tangent at M intercept between M and the y-axis is shortest.**

Sol. We have

$$y = \frac{3}{\sqrt{2}} x \ln x, x \in \left(e^{-1.5}, \infty\right)$$

And $y' = \frac{3}{\sqrt{2}}(1+\ln x), x \in \left(e^{-1.5}, \infty\right)$ ...(i)

Any point on the curve can be chosen as $M \equiv \left(h, \frac{3}{\sqrt{2}} h \ln h\right)$

Slope of the tangent at $M = \frac{3}{\sqrt{2}}(1+\ln h)$ [using eq.(i)]

Equation of the tangent at M, is given by

$$y - \frac{3}{\sqrt{2}} h \ln h = \frac{3}{\sqrt{2}}(1+\ln h)(x-h) \quad \text{...(ii)}$$

The tangent cuts the Y-axis at

$N \equiv \left(0, \frac{-3h}{\sqrt{2}}\right)$ [putting x = 0 in eq. (ii)]

Length l of the tangent segment MN, is given by

$$l^2 = h^2 + \left\{\frac{3}{\sqrt{2}} h(1+\ln h)\right\}^2 = h^2 + \frac{9}{2} h^2 (1+\ln h)^2$$

The value of h at which l^2 attains minima, is given by the equation

$$\frac{d}{dh}\left(l^2\right) = 0$$

i.e. $2h + 9h(1+\ln h)^2 + 9h^2(1+\ln h)\left(\frac{1}{h}\right) = 0$

i.e. $h\left[(1+\ln h)^2 + (1+\ln h) + \frac{2}{9}\right] = 0$

i.e. $1+\ln h = \frac{-1 \pm \sqrt{1-\frac{8}{9}}}{2} = \frac{-2}{3}, \frac{-1}{3}$ $\left[h = 0 \text{ does not lie in } \left(e^{-1.5}, \infty\right)\right]$

gives $h = e^{-\frac{5}{3}}, e^{-\frac{4}{3}}$.

Only $h = e^{\frac{-4}{3}}$ is acceptable since $e^{\frac{-5}{3}}$ does not lie in $\left(e^{-1.5}, \infty\right)$.

Hence, the required point, is $M \equiv \left(e^{\frac{-4}{3}}, -2\sqrt{2}e^{\frac{-4}{3}}\right)$

Q.4 **Assuming that the petrol burnt per unit time in driving a motor boat, varies as the cube of its velocity. Find the most economical speed of boat when moving against a current whose speed is 'c' m/sec.**

Sol. Let s m/sec be the speed of the motor boat in still water. Then, the rate at which petrol is burnt, is given by

$$\frac{dp}{dt} = ks^3 \qquad \ldots(1)$$

where k is a constant of proportionality.
Now that the boat is moving against a current of 'c' m/sec, the absolute speed of the motor boat is (s-c)m/sec. Then, the rate at which distance is covered by the motor boat, is given by

$$\frac{dx}{dt} = s - c \qquad \ldots(2)$$

Dividing equation (1) by equation (2)

$$\frac{dp}{dx} = \frac{ks^3}{s-c}$$

which is the amount of petrol consumed per unit distance travelled by the boat..

$$f(s) = \frac{ks^3}{s-c}$$

The value of s at which f(s) attains minima, is given by

$$\frac{df}{ds} = 0$$

i.e. $$\frac{(s-c)3s^2 - s^3}{(s-c)^2} = 0$$

i.e. $2s - 3c = 0$ given $s = \frac{3}{2}c$

At this value of s, the petrol consumption per unit distance, is minimum and the absolute speed of the motor boat is

$$\frac{3}{2}c - c = \frac{c}{2} m/sec.$$

Q.5 **Find the point on the curve $y = \frac{x}{1+x^2}$ where the tangent to the curve has the greatest slope.**

Sol. We have

$$y = \frac{x}{1+x^2}$$

The slope of the tangent at any point on the curve, is given by

$$\frac{dy}{dx} = \frac{\left(1+x^2\right)1 - x\left(2x\right)}{\left(1+x^2\right)^2} = \frac{1-x^2}{\left(1+x^2\right)^2}$$

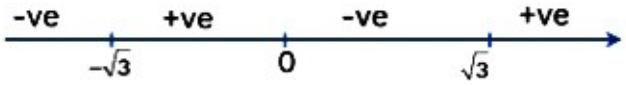
The values at which the slope becomes maximum or minimum, is given by

$$\frac{d^2y}{dx^2} = 0$$

i.e. $$\frac{d}{dx}\left[\frac{1-x^2}{\left(1+x^2\right)^2}\right] = 0$$

i.e. $$\frac{\left(1+x^2\right)^2\left(-2x\right) - \left(1-x^2\right)2\left(1+x^2\right).2x}{\left(1+x^2\right)^4} = 0$$

i.e. $$\frac{2x\left(1+x^2\right)\left(x^2-3\right)}{\left(1+x^2\right)^4} = 0$$



i.e. $x = 0, \pm\sqrt{3}$.

Now, from the sign scheme for $\frac{d^2y}{dx^2}$, we have

$\frac{d^2y}{dx^2}$ changes sign from +ve to -ve at $x = 0$

$\Rightarrow \frac{dy}{dx}$ is a maximum at $x = 0$

Hence, the point on the curve where slope of the tangent is maximum, is (0, 0).

Q.6 **Discuss the extremum of f(x) = a sec x + b cosec x, 0 < a < b.**

Sol. $f(x)=a\sec x + b\,\text{cosec}\,x, 0<a<b$.

$f'(x)=a\sec x\tan x - b\,\text{cosec}\,x\cot x$

Let $f'(x)=0 \Rightarrow a\dfrac{\sin x}{\cos^2 x} - b\dfrac{\cos x}{\sin^2 x}=0$

$\Rightarrow \tan^3 x = \dfrac{b}{a} \Rightarrow x = \tan^{-1}\left(\dfrac{b}{a}\right)^{\frac{1}{3}}; a,b>0 \Rightarrow x = \tan^{-1}\left(\dfrac{b}{a}\right)^{\frac{1}{3}} > 0$

$\Rightarrow$ x lies in either the first or third quadrant for extremum.

Case-I : $0 < x < \dfrac{\pi}{2}$

$\lim\limits_{x\to 0}(a\sec x + b\,\text{cosec}\,x) \to \infty$

$\lim\limits_{x\to \frac{\pi}{2}}(a\sec x + b\,\text{cosec}\,x) \to \infty$

Also f(x) is +ve for this value of x.
Hence, only one point of extremum is the point of minima.

and $\tan x = \left(\dfrac{b}{a}\right)^{\frac{1}{3}}$

$\Rightarrow \cos x = \dfrac{a^{\frac{1}{3}}}{\sqrt{a^{\frac{2}{3}}+b^{\frac{2}{3}}}}, \sin x = \dfrac{b^{\frac{1}{3}}}{\sqrt{a^{\frac{2}{3}}+b^{\frac{2}{3}}}}$

$\Rightarrow$ minimum value of $f = \dfrac{a\sqrt{a^{\frac{2}{3}}+b^{\frac{2}{3}}}}{a^{\frac{1}{3}}} + \dfrac{b\sqrt{a^{\frac{2}{3}}+b^{\frac{2}{3}}}}{b^{\frac{1}{3}}} = \left(a^{\frac{2}{3}}+b^{\frac{2}{3}}\right)^{\frac{3}{2}}$

Case-II : $\pi < x < \dfrac{3\pi}{2}$

$\lim\limits_{x\to \pi}(a\sec x + b\,\text{cosec}\,x) \to -\infty$

$\lim\limits_{x\to \frac{3\pi}{2}}(a\sec x + b\,\text{cosec}\,x) \to -\infty$

Also f(x) is -ve for this value of x.
Hence, only one point of extremum is the point of maximum.

$\Rightarrow$ Maximum value $f_{max} = -\left(a^{\frac{2}{3}}+b^{\frac{2}{3}}\right)^{\frac{3}{2}}$ (Local Maxima)

Q.7 **A running track of 440 ft is to be laid out enclosing a football field, the shape of which is a rectangle with a semi-circle at each end. If the area of the rectangular portion is to be maximum, then find the length of its sides.**

Sol. Perimeter = 440 ft

$\Rightarrow 2x + \pi r + \pi r = 440$ or $2x + 2\pi r = 440$

A= Area of the rectangular portion =x2r

$$\Rightarrow A = \frac{(440 - 2x)x}{\pi} = \frac{1}{\pi}\left(440x - 2x^2\right)$$

$$\text{Let } \frac{dA}{dx} = \frac{1}{\pi}(440 - 4x) = 0$$

$$\Rightarrow x = 110 \text{ for which } \frac{d^2A}{dx^2} < 0$$

$\Rightarrow$ A is maximum when x = 110

$$\Rightarrow 2r = \frac{440 - 220}{\frac{22}{7}} = 70$$

$\Rightarrow r = 35$ ft and $x = 110$ ft

Q.8 **Find the point on the curves $5x^2 - 8xy + 5y^2 = 4$ whose distance from the origin is maximum or minimum.**

Sol. Let(r, θ)be the polar coordinates of any point P on the curve where r is the distance of the point from the origin.

$$\Rightarrow r^2\left[5\left(\cos^2\theta + \sin^2\theta\right) - 8\sin\theta\cos\theta\right] = 4$$

$$\Rightarrow r^2 = \frac{4}{5 - 4\sin 2\theta}$$

r^2 is maximum when $5 - 4\sin 2\theta$ is minimum = 5 − 4 = 1 (when $\sin 2\theta = 1$)

$\Rightarrow 2\theta = 90^\circ \Rightarrow \theta = 45^\circ \Rightarrow r = \pm 2, \theta = 45^\circ$ ...(i)

Again r^2 is minimum when $5 - 4\sin 2\theta$ is maximum

$$= 5 + 4 = 9 \text{ when } \sin 2\theta = -1 \Rightarrow 2\theta = \frac{3\pi}{2} \Rightarrow \theta = \frac{3\pi}{4}$$

$$\Rightarrow r = \pm\frac{2}{3}, \theta = \frac{3\pi}{4}$$ ...(ii)

Hence, the point are $(r\cos\theta, r\sin\theta)$ where r and θ are given by equation (i) and (ii).

Thus, we get four point $(\sqrt{2},\sqrt{2}),(-\sqrt{2},-\sqrt{2}),\left(\frac{\sqrt{2}}{3},-\frac{\sqrt{2}}{3}\right)$ and $\left(-\frac{\sqrt{2}}{3},\frac{\sqrt{2}}{3}\right)$.

Q.9 **Discuss the monotonocity of Q(x), where** $Q(x)=2f\left(\frac{x^2}{2}\right)+f\left(6-x^2\right)\forall x\in R$**. It is given that** $f''(x)>0\forall x\in R$**. Find also the point of maxima and minima of Q(x).**

Sol. Given $Q(x)=2f\left(\frac{x^2}{2}\right)+f\left(6-x^2\right)$

$$\therefore Q'(x)=2xf'\left(\frac{x^2}{2}\right)-2xf'\left(6-x^2\right)=2x\left\{f'\left(\frac{x^2}{2}\right)-f'\left(6-x^2\right)\right\}$$

But given that $f''(x)>0\Rightarrow f'(x)$ is increasing for all $x\in R$.

Case-I : Let $\frac{x^2}{2}>\left(6-x^2\right)\Rightarrow x^2>4$

$\therefore x\in(-\infty,-2)\cup(2,\infty)$

$\Rightarrow f'\left(\frac{x^2}{2}\right)>f'\left(6-x^2\right)\forall x\in(-\infty,-2)\cup(2,\infty)$

$\Rightarrow f'\left(\frac{x^2}{2}\right)-f'\left(6-x^2\right)<0\forall x\in(-2,2)$

$\Rightarrow Q'(x)<0 \qquad \Rightarrow x\in(-\infty,-2)\cup(0,2)$

$\Rightarrow Q'(x)>0\Rightarrow x\in(-2,0)\cup(2,\infty)$

There is a maxima at x = 0 and minima at x = –2 and x = 2

Q.10 **The largest term in the sequence** $a_n=\frac{n^2}{n^3+200}$ **is given by**

(A) $\frac{529}{49}$ **(B)** $\frac{8}{89}$ **(C)** $\frac{49}{543}$ **(D) None of these**

Sol. Consider the function $f(x) = \dfrac{x^2}{(x^3 + 200)}$ ...(i)

$$f'(x) = x\frac{(400 - x^3)}{(x^3 + 200)^2} = 0$$

when $x = (400)^{\frac{1}{3}} \quad (\because x \neq 0)$

$x = (400)^{\frac{1}{3}} - h \Rightarrow f'(x) > 0$

$x = (400)^{\frac{1}{3}} + h \Rightarrow f'(x) < 0$

$\therefore f(x)$ has maxima at $x = (400)^{\frac{1}{3}}$

Since $7 < (400)^{\frac{1}{3}} < 8$, eithe a_7 or a_8 is the greatest term of the sequence.

$\because a_7 = \dfrac{49}{543}$ and $a_8 = \dfrac{8}{89}$ and $\dfrac{49}{543} > \dfrac{8}{89}$

$\Rightarrow a_7 = \dfrac{49}{543}$ is the greatest term.

Q.11 **Prove the inequality :** $\sin x + 2x \geq \dfrac{3x(x+1)}{\pi} \forall x \in \left[0, \dfrac{\pi}{2}\right]$

Sol. Let $f(x) = \sin x + 2x$ and $g(x) = \dfrac{3x(x+1)}{\pi}$

$f'(x) = \cos x + 2 > 0 \forall x \in \left[0, \dfrac{\pi}{2}\right]$

$f''(x) = -\sin x < 0 \; \forall \; x \in \left(0, \dfrac{\pi}{2}\right)$

So, f(x) will be increasing but will have its graph concave downwards.

$g'(x) = \dfrac{6x + 3}{\pi} > 0 \; \forall \; x \in \left[0, \dfrac{\pi}{2}\right]$

$g''(x) = \dfrac{6}{\pi} > 0$

$\therefore$ g(x) will be increasing and its graph will be concave upwards.

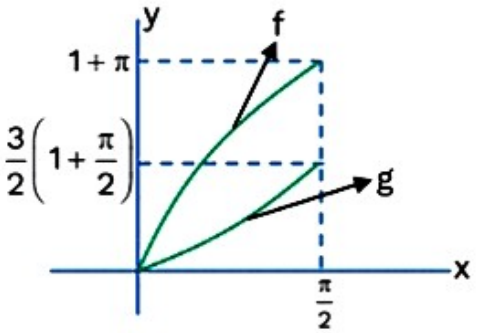
$$f(0) = 0 \; ; f\left(\frac{\pi}{2}\right) = 1 + \pi \approx 4.1$$

$$g(0) = 0 \; ; g\left(\frac{\pi}{2}\right) = \frac{3}{2}\left(1 + \frac{\pi}{2}\right) < 4$$

$$\Rightarrow f\left(\frac{\pi}{2}\right) > g\left(\frac{\pi}{2}\right)$$

From above graph, it can be seen that

$$f(x) \geq g(x) \forall x \in \left[0, \frac{\pi}{2}\right]$$



Hence proved.